COUNTRY PROFILES

Russia's Nuclear Fuel Cycle

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- A significant increase in uranium mine production is planned.
- There is increasing international involvement in parts of Russia's fuel cycle.
- A major Russian political and economic objective is to increase exports, particularly for front-end fuel cycle services through Tenex, as well as nuclear power plants.

Russia uses about 5500 tonnes of natural uranium per year.

Privacy - Terms

There is high-level concern about the development of new uranium deposits, and a Federal Council meeting in April 2015 agreed to continue the federal financing of exploration and estimation works in Vitimsky Uranium Region in Buryatia. It also agreed to financing construction of the engineering infrastructure of Mine No. 6 of Priargunsky Industrial Mining and Chemical Union (PIMCU). The following month the Council approved key support measures including the introduction of a zero rate for mining tax and property tax; simplification of the system of granting subsoil use rights; inclusion of the *Economic Development of the Far East and Trans-Baikal up to 2018* policy in the Federal Target Program; and the development of infrastructure in Krasnokamensk.

In June 2015 Rosgeologia signed a number of agreements to expedite mineral exploration in Russia, including one with Rosatom. It was established in July 2011 by presidential decree and consists of 38 enterprises located in 30 regions across Russia, but uranium is a minor part of its interests.

Uranium resources and mining

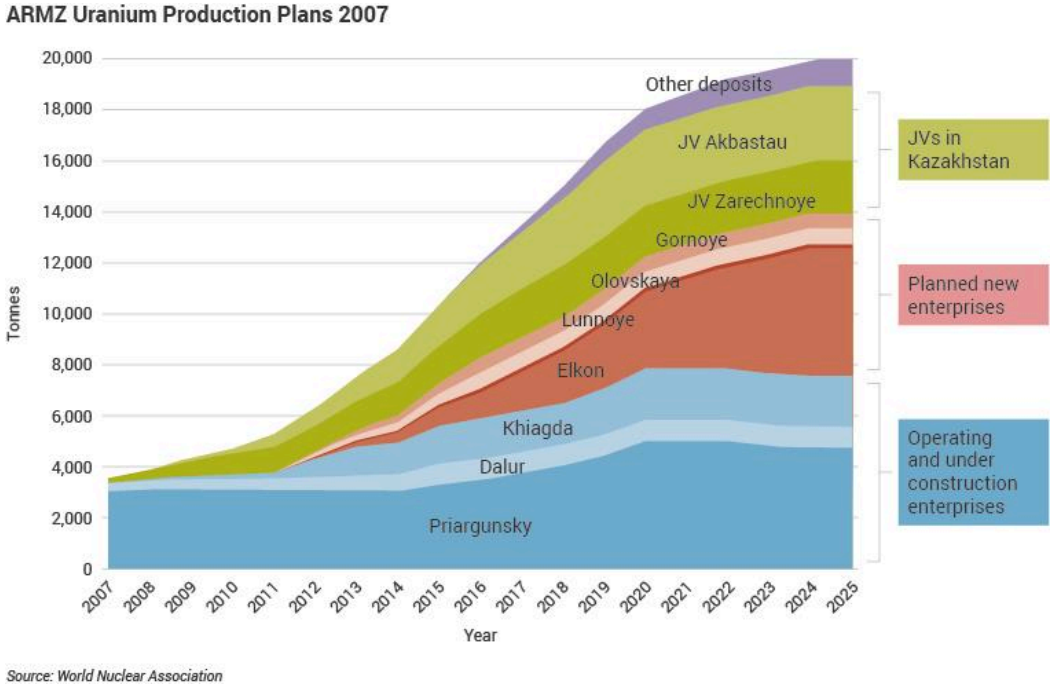
Russia has substantial economic resources of uranium, with about 9% of world reasonably assured resources plus inferred resources up to $130/kg – 505,900 tonnes U (2014 *Red Book*). Rosatom reported ARMZ resources as 517,000 tU in September 2015, mostly requiring underground mining. Historic uranium exploration expenditure is reported to have been about $4 billion. The Federal Natural Resources Management Agency (Rosnedra) reported that Russian uranium reserves grew by 15% in 2009, particularly through exploration in the Urals and Kalmykia Republic, north of the Caspian Sea.

Uranium production has varied from 2870 to 3560 tU/yr since 2004, and in recent years has been supplemented by that from Uranium One Kazakh operations, giving 7629 tU in 2012. In 2006 there were three mining projects in Russia, since then others have been under construction and more projected, as described below. Cost of production in remote areas such as Elkon is said to be US$ 60-90/kg. Spending on new ARMZ domestic projects in 2013 was RUR 253.5 million, though in November 2013 all Rosatom investment in mining expansion was put on hold due to low uranium prices.

Plans announced in 2006 for 28,600 t/yr U_3O_8 output by 2020, 18,000t of this from Russia* and the balance from Kazakhstan, Ukraine, Uzbekistan and Mongolia have since taken shape, though

difficulties in starting new Siberian mines makes the 18,000 t target unlikely. Three uranium mining joint ventures were established in Kazakhstan with the intention of providing 6000 tU/yr for Russia from 2007: JV Karatau, JV Zarechnoye and JV Akbastau (see below and Kazakhstan paper).

* See details for April 2008 ARMZ plans. In 2007 TVEL applied for the Istochnoye, Kolichkanskoye, Dybrynskoye, Namarusskoye and Koretkondinskoye deposits with 30,000 tU in proved and probable reserves close to the Khiagda mine in Buryatia.
From foreign projects: Zarechnoye 1000 t, Southern Zarechnoye 1000 t, Akbastau 3000 t (all in Kazakhstan); Aktau (Uzbekistan) 500 t, Novo-Konstantinovskoye (Ukraine) 2500 t. In addition Russia would like to participate in development of Erdes deposit in Mongolia (500t) as well as in Northern Kazakhstan deposits Semizbai (Akmolonsk Region) and Kosachinoye.



*(this chart is now slightly out of date but still gives a general picture)

AtomRedMetZoloto (ARMZ) is the state-owned company which took over Tenex and TVEL uranium exploration and mining assets in 2007-08, as a subsidiary of Atomenergoprom (79.5% owned). It inherited 19 projects with a total uranium resource of about 400,000 tonnes, of which 340,000 tonnes are in Elkonskiy uranium region and 60,000 tonnes in Streltsovskiy and Vitimskiy regions. The rights to all these resources had been transferred from Rosnedra, the Federal Agency for Subsoil Use under the Ministry of Natural Resources and Environment.

JSC ARMZ Uranium Holding Company (as it is now known) became the mining division of Rosatom in 2008, responsible for all Russian uranium mine assets and also Russian shares in foreign joint ventures. In 2008, 78.6% of JSC Priargunsky, all of JSC Khiagda and 97.85% of JSC Dalur was transferred to ARMZ. In March 2009 the Federal Financial Markets Service of Russia registered RUR 16.4 billion of additional shares in ARMZ placed through a closed subscription to pay for uranium mining assets, on top of a RUR 4 billion issued in mid 2008 to pay for the acquisition of Priargunsky, Khiagda and Dalur. In November 2009 SC Rosatom paid a further RUR 33 billion for ARMZ shares, increasing its equity to 76.1%.

In 2009 and 2010 ARMZ took a 51% share in Canadian-based Uranium One Inc, paying for this with $610 million in cash and by exchange of assets in Kazakhstan: 50% of JVs Akbastau, Karatau and Zarechnoye, mining the Budenovskoye and Zarechnoye deposits. (An independent financial advisor put the value of ARMZ's stakes in the Akbastau and Zarechnoye JVs at $907.5 million.) Uranium One has substantial production capacity in Kazakhstan, including now those two mines with Karatau, Akdala, South Inkai and Kharasan, as well as small prospects in USA and Australia (sold in 2015). In 2013 ARMZ completed the purchase of outstanding shares in Uranium One Inc, and it became a full subsidiary of ARMZ. JSC Uranium One Group (U1 Group) is from December 2016 a 78.4% owned subsidiary of Atomenergoprom and apparently separate from ARMZ.

Following this, late in 2013 Rosatom established Uranium One Holding NV (U1H) as its global growth platform for all international uranium mining assets belonging to Russia, with headquarters in Amsterdam. It lists assets in Kazakhstan, USA and Tanzania, as well as owning and managing Rosatom's stake in Uranium One Inc. In 2013 it accounted for 5086 tU production at average cash cost of $16/lb U_3O_8, and reported 229,453 tU measured, indicated and inferred resources (attributable share). In 2014 it produced 4857 tU and listed resources of 177,000 tU. The company plans to extend its interests into rare earths. Its 'strategic partner' is JSC NAC Kazatomprom.

ARMZ remains responsible for uranium mining in Russia. At the end of 2013 it was 82.75% owned by Rosatom and 17.25% TVEL. Exploration expenditure has nearly doubled in two years to about US$ 52 million in 2008. In 2013 the government approved an exploration budget of RUR 14 billion ($450 million) through to 2020, principally in the Far East and Northern Siberia. Deposits suitable for ISL mining will be sought in the Transurals, Transbaikal and Kalmykyia. Other work will be in the Urals, Siberian, Far East

Federal Districts (Zauralsky, Streltsovsky, Vitimsky and Vostochno-Zabaikalsky, and Elkonsky ore regions).

Rosgeologia, the Russian state-run geological exploration services company set up in 2011, has identified "promising" uranium deposits in the North-West Federal District of Russia following completion of a survey of the Kuol-Panayarvinskaya area on the border of the Murmansk region and the Republic of Karelia. It signed an agreement with Rosatom in 2015 to focus on uranium.

CJSC Rusburmash (RBM) is the exploration subsidiary of ARMZ. VNIPIPT is the subsidiary responsible for R&D and engineering of mining and processing plants.

In December 2010 ARMZ made a $1.16 billion takeover bid for Australia's Mantra Resources Ltd with a prospective Mkuju River project in southern Tanzania, which was expected in production about 2013 at 1400 tU/yr, but is now deferred. This is now under U1H.

Domestic mining

In 2009 the government accepted Rosatom's proposal for ARMZ and Elkonsky Mining and Metallurgical Combine to set up the "open-type joint stock company" EGMK-Project. The state's contribution through Rosatom to the EGMK-Project authorized capital will be RUR 2.657 billion, including RUR 2.391 billion in 2009 and RUR 0.266 billion in 2010. EGMK-Project is being set up to draw up the project and design documentation for Elkonsky Mining and Metallurgical Combine (see below).

The Russian Federation's main uranium deposits are in four districts:

- The Trans-Ural district in the Kurgan region between Chelyabinsk and Omsk, with the Dalur ISL mine.
- Streltsovskiy district in the Transbaikal or Chita region of SE Siberia near the Chinese and Mongolian borders, served by Krasnokamensk and with major underground mines.
- The Vitimsky district in Buryatia about 570 km northwest of Krasnokamensk, with the Khiagda ISL mine.
- The more recently discovered remote Elkon district in the Sakha Republic (Yakutia) some 1200 km north-northeast of the Chita

region.

Present production by ARMZ is principally from the Streltsovskiy district, where major uranium deposits were discovered in 1967, leading to large-scale mining, originally with few environmental controls. These are volcanogenic caldera-related deposits. Krasnokamensk is the main town serving the mines.

In 2008 ARMZ said that it intended to triple production to 10,300 tU per year by 2015, with some help from Cameco, Mitsui and local investors. ARMZ planned to invest RUR 203 billion (US$ 6.1billion) in the development of uranium mining in Russia in 2008-2015. It aimed for 20,000 tU per year by 2024. Total cost was projected at RUR 67 billion ($2 billion), mostly at Priargunsky, with RUR 4.8 billion ($144 million) there by end of 2009 including a new $30 million, 500 tonne per day sulfuric acid plant commissioned in 2009, replacing a 1976 acid plant.

Russian uranium mining

Production centre	Region	First production	Orebody	Known resources: tU	Capacity: tU/yr
Priargunsky	Transbaikal/Chita	1968	volcanic	95,700 @ 0.16%	3000
Dalur	Trans-Ural/Kurgan	2004	sandstone	7,400 @ 0.04%	700
Khiagda	Buryatia, Vitimsky	2010	sandstone	29,900 @ 0.05%	1000
Gornoye	Transbaikal/Chita	deferred	granite, vein	3,200 @ 0.20%	300
Olovskaya	Transbaikal/Chita	deferred	volcanic	8,210 @ 0.072%	600
Elkon	Yakutia/Sakha	(2020)	metasomatite	303,600 @ 0.15%	5000
Lunnoye	Yakutia/Sakha	(2016?)	polymetallic	800	100 with gold

Source: 2016 'Red Book' except Olovskaya and Lunnoye.

Russian uranium production, tonnes U

Production centre	2012	2013	2014	2015	2016	2017	2018	2019	2020
Priargunsky	2011	2133	1970	1977	1873	1631	1456	1300	1240

Production centre	2012	2013	2014	2015	2016	2017	2018	2019	2020
Dalur	529	562	578	590	591	592	858	595	585
Khiagda	332	440	442	488	540	693	590	1016	1021
Total	2872	3135	2990	3055	3004	2916	2904	2911	2846

Trans-Ural, Kurgan region

A modest level of production is from **Dalur** in the Trans-Ural Kurgan region. This is a low-cost ($40/kg) acid in situ leach (ISL) operation in sandstones. About 1350 km east of Moscow, Uksyanskoye is the town supporting the Dalur mine. ARMZ's 2008 plan had production at Dalur by acid ISL increasing from 350 to 800 tU/yr by 2019 (expanding from the Dalmatovskoye field in the Zauralsk uranium district to Khokhlovskoye in the Shumikhinsky district, then Dobrovolnoye in the Zverinogolovsky district). In 2014 JSC Dalur completed further exploration of the Khokhlovskoye deposit and increased its resources from 4700 to 5500 tonnes. A mill upgrade was started in 2016. More than half of 2016 production was from the Ust-Uksyansky part of Dalmatovskoye field.

In 2016 geological exploration at the Dobrovolnoye deposit was advanced, and a permit for development was received in June 2017, allowing construction of the pilot plant, which commenced in 2020. Its reserves are quoted as 7067 tU. After pilot operation to 2021, commercial operation is expected to maintain Dalur production at 700 tU per year to about 2025 after Dalmatovskoye and Khokhlovskoye are exhausted, reaching full capacity in 2031.

Transbaikal Chita region, Streltsovskiy district

Here, several underground mines operated by JSC **Priargunsky** Industrial Mining and Chemical Union (PIMCU – 85% ARMZ) supply low-grade ore to a central mill near Krasnokamensk. PIMCU was established in 1968, and produces some other metals than uranium. Since 2008 it has been an ARMZ subsidiary. Historical production from Priargunsky is reported to be 140,000 tU (some from open cut mines) and 2011 known resources (RAR + IR) are quoted as 115,000 tU at 0.159%U. In 2013 'reserves' were quoted by ARMZ at 108,700 tonnes. Production is up to about 3000 tU/yr, about one-tenth of it from heap leaching. In 2015 production was

1977 tU and costs were reduced by 11%, so that it hoped to break even in mid-2016.

The company has six underground mines, most of them operating: Mine #1, Mine #2, Glubokiy Mine, Shakhta 6R, Mine #8 with extraction from Maly Tulukui deposit, and Mine #6 (see below). ARMZ's 2008 plan called for Priargunsky's production to be expanded from 3000 to 5000 tU/yr by 2020.

Mine #1 production rate was increased in 2016. It is on the opposite side of the Oktyabriski settlement from mine #2 and about 2 km from it.

Mine #2 was making a loss in 2013 due to market conditions, so it was closed in order to concentrate on bringing mine #8 to full production. Stoping operations resumed in February 2015, with production target 130 tU for the year, from average grade 0.15%. It is now known as section 2 of mine #8. Some production has been exported to France, Sweden and Spain.

Mine #8 began producing in 2011, towards phase 1 target capacity of 400 t/yr by the end of 2014. The total cost of development is expected to be RUR 4.8 billion (RUR 3.5 billion for phase 1). Production was increased 22% in 2016.

Mine #6 will access the Argunskoye and Zherlovoye deposits which comprise 35% of the Streltsovskoye reserves of 40,900 tU, with much higher grade (0.3%U) than the rest. Production cost from mine #6 is projected at $90/kgU. Future plans for Priargunsky are focused on development of mine #6, official construction of which commenced in 2018.

Development began in 2009 for stage 1 production from 2015 to reach full capacity in 2019, but this was put on hold in 2013. In March 2015 ARMZ said it hoped to find co-investors in the project, and federal funds might be forthcoming. Then in June 2015 Rosatom's Investment Committee decided to finance the development. In August 2016 ARMZ said that RUR 27 billion was required to enable 2022 commissioning. In March 2018 a new financing arrangement was announced to the extent of RUR 18.5 billion, with Priargunsky to own 51% of the project and ARMZ 49% directly. Most of the project financing – RUR 16.1 billion – would be from China National Nuclear Corporation (CNNC), with the balance of RUR 2.5 billion from a new Russia-China Investment Fund for Regional Development (RCIF) "as a first step in widening cooperation" with China. According to the *Russian Gazette* (quoted by Platts *Nuclear Fuel*), CNNC's investment would give it a 49% stake in the joint venture, entitling it to that proportion of annual

production. Construction recommenced in March 2018, aiming for first production in 2023, ramping up to full capacity of 1800 tU/yr by 2026. Rosatom reported that the Mine #6 development project is supervised by the government of Zabaikalsky Krai.

Mine #4. Mining the Tulukuy pit of Mine #4 ceased in 1991 due to low grades, but now low-cost block-type underground leaching is ready to be employed in the pit bottom to recover the remaining 6000 tU. Following this the pit will be filled with low-grade ore for heap leaching.

A re-evaluation of reserves in 2012 suggested that mineable resources apart from Mine #6 amounted to only 32,000 tU. Mine #8 resources were quoted at 12,800 tU in December 2012. In 2014 PIMCU, as part of the Kaldera project, identified four promising areas over 100 sq km in the Streltsovskoye ore field, with resources estimated at 80,000 tU, and they will be explored over 2015-17.

In 2014 PIMCU completed an upgrade of its sulfuric acid plant to take daily production from 400 to 500 tonnes, for use in both the conventional mill and in underground and heap leaching. Also the mill (hydrometallurgical plant) process was improved.

There is a legacy environmental problem at Priargunsky arising from 30 waste rock and low-grade ore dumps as well as tailings. Rehabilitation of waste rock dumps and open pits is proceeding and low-grade ores are being heap leached. Dams and intercepting wells below the tailings dams with hydrogeological monitoring and wastewater treatment is addressing water pollution. Final rehabilitation of the impacted areas will occur after final closure takes place. In 2016 ARMZ announced a new heap leaching initiative for very low-grade ores stockpiled on the surface, to produce 50 to 63 tU/yr.

In 2006 Priargunsky won a tender to develop Argunskoye and Zherlovoye deposits in the Chita region with about 40,000 tU reserves. Dolmatovsk and Khokhlovsk have also been identified as new mines to be developed (location uncertain).

Development of **Olovskoye and Gornoye** deposits* in the Transbaikal region near Priargunsky towards Khiagda would add 900 tU/yr production for RUR 135 billion ($5.7 billion). Measured resources together are 12,200 tU and inferred resources 1600 tU, all at 0.072% average (JORC-compliant). In 2007 newly-formed ARMZ set up two companies to undertake this, and possibly attract some foreign investment:

- Gornoye Uranium Mining Company (UDK Gornoye) to develop the Gornoye and Berezovoye mines in the Krasnochikoysky and Uletovsky districts in Chita, with underground mining and some heap leach (ore grade 0.226%U) originally to produce 300 tU/yr from 2014, but now anticipating up to 1000 tU/yr from 2025.

- Olovskaya Mining & Chemical Company to develop the Olovskoye deposits in the Chernyshevsk district of Chita region with underground, open cut and heap leach to produce 600 tU/yr from 2016.

The 2016 *Red Book* noted that UDK Gornoye was undertaking pilot mining project design for the Berezovoye deposit.

* 2006 plans were for 2000t/yr at new prospects in Chita Region and Buryatia (Gornoye, Berezovoye, Olovskoye, Talakanskoye properties etc.), plus some 3000t at new deposits.

Buryatia, Vitimsky district

JSC **Khiagda**'s operations are at **Vitimsky** in Buryatia about 570 km northwest of Krasnokamensk, serving Priargunsky's operations in Chita region, and 140 km north of Chita city. They are starting from a low base – in 2010 production from the Khiagdinskoye ore field was 135 tU, rising to 440 tU in 2013 (fully utilising the pilot plant) and targeting 1000 tU/yr from 2018 with a new plant. These are a low-cost (US$ 70/kgU) acid in situ leach (ISL) operations in sandstones, and comprise the only ISL mine in the world in permafrost. Groundwater temperature is 1-4°C, giving viscosity problems, especially when winter air temperature is -40°C. The main uranium mineralisation is a phosphate, requiring oxidant addition to the acid solution. In the Khiagdinskoye field itself there are eight palaeochannel deposits over 15 x 8 km, at depths of 90 to 280 metres (average 170 m). Single orebodies are up to 4 km long and 15 to 400 m wide, 1 to 20 m thick.

JSC Khiagda has resources of 55,000 tU amenable to ISL mining, with resource potential estimated by Rosatom of 350,000 tU, giving a mine life of over 50 years. In 2015 'reserves' were quoted by ARMZ at 39,300 tonnes U. The 2008 ARMZ plan envisaged production from JSC Khiagda's project increasing to 1800 tU/yr by 2019, but in 2013 the higher target was postponed. The 2018 plan is now 1000 tonnes. In 2014 JSC Khiagda continued construction of the main production facility and on the sulfuric acid plant, the first

stage of which was commissioned in September 2015. Its final design capacity is 110,000 t/yr.

JSC Khiagda is currently mining uranium from the Khiagdin and Istochnoy deposits of the Khiagda ore field. Preparatory work for mining operations at the Vershinny deposit is under way. In May 2018, JSC Khiagda announced that engineering and geological surveys ahead of the construction of mining facilities was under way at Kolichikan and Dybryn deposits. The other two fields in the immediate vicinity are Namaru and Tetrakhskoye. All these deposits occur over an area about 50 x 20 km. There are also plans to install plant for extracting rare earth oxides (REO) as by-product. The nearest towns are Romanovka, 133 km north of Chita, and Bagdarin.

Sakha/Yakutia, Elkon district

ARMZ's long-term hope is development of the massive **Elkon** project with several mines in the Sakha Republic (Yakutia) some 1200 km north-northeast of the Chita region. The Elkon project is in a mountainous region with difficult climate conditions and little infrastructure, making it a challenging undertaking. Production from metasomatite deposits is planned to ramp up to 5000 tU/yr over ten years, for RUR 90.5 billion ($3 billion), and 2020 start up was envisaged, but this is now "after 2030". Elkon is set to become Russia's largest uranium mining complex, based on resources of over 270,000 tU (or 357,000 tU quoted by Rosatom in 2015). It will involve underground mining, radiometric sorting, milling, processing and uranium concentrate production of up to 5000 tU/yr.

Elkon Mining and Metallurgical Combine (EMMC) was set up by ARMZ to develop the substantial Elkonsky deposits. The Elkon MMC project involves the JSC Development Corporation of South Yakutia and aims to attract outside funding to develop infrastructure and mining in a public-private partnership, with ARMZ holding 51%. Foreign equity including from Japan, South Korea and India is envisaged, and in March a joint venture arrangement with India was announced. The Elkon MMC developments are to become "the locomotive of the economic development of the entire region", building the infrastructure, electricity transmission lines, roads and railways, as well as industrial facilities, from 2010. Of 15 proposed construction sites, three have been tentatively selected: at the mouth of Anbar River, Diksi Village and Ust-Uga Village. The building of four small floating co-generation plants to supply heat

and electricity to northern regions of Yakutia is linked with the Elkon project in southern Yakutia.

There are eight deposits in the Elkon project with resources of 320,000 tU* (RAR + IR) at average 0.146%U, with gold by-product: Elkon, Elkon Plateau, Kurung, Neprokhodimoye, Druzhnoye (southern deposits), as well as Yuzhnaya, Severnaya, Zona Interesnaya and Lunnoye (see below). In mid-2010 ARMZ released JORC-compliant resource figures for the five southern deposits: 71,300 tU as measured and indicated resources, and 158,500 tU as inferred resources, averaging 0.143%U. ARMZ pointed out that the resource assessment against international standards will increase the investment attractiveness of EMMC. However, in September 2011 ARMZ said that production costs would be US$ 120-130/kgU, which would be insufficient in the current market, and costs would need to be cut by 15-20%.

* 257,800 tU of this was in the five southern deposits. The 2011 *Red Book* gives 271,000 tU resources for Elkon, or 319,000 tU in situ.

First production from EMMC was expected in 2015 ramping up to 1000 tU/yr in 2018, 2000 tU/yr in 2020 and 5000 tU/yr by 2024 based on the southern deposits as well as Severnoye and Zona Interesnoye. This schedule has slipped by at least ten years. Also, it is remote, and mining will be underground, incurring significant development costs. ARMZ and EMMC are seeking local government (Sakha) support for construction of main roads and railways to access the Elkon area, and make investment there more attractive.

JSC **Lunnoye** was set up by ARMZ at the same time as EMMC to develop a small deposit jointly by ARMZ (50.1%) and a gold mining company Zoloto Seligdara as a pilot project to gain practical experience in the region in a polymetallic orebody. Lunnoye is expected in full production in 2016, reaching 100 tU/yr. It has reserves of 800 tU and 13 t gold, and is managed by Zoloto Seligdara. ARMZ in mid 2011 expressed impatience with the rate of development.

Further mine prospects

The Federal Subsoil Resources Management Agency (Rosnedra) was transferring about 100,000 tonnes of uranium resources to miners, notably ARMZ, in 2009-10, and 14 projects, mainly small to medium deposits, were prepared for licensing then. They are located mainly in the Chita (Streltsovskiy district), Trans-Ural

(Zauralskiy district) and Buryatia (Vitimskiy district) uranium regions.

The projects prepared for licensing include:

- Chita Oblast – Zherlovskoye, Pyatiletnee, Dalnee and Durulguevskoye.
- Republic of Buratiya – Talakanskoye, Vitlausskoye, Imskoye, Tetrakhskoye, and Dzhilindinskoye.
- Kurgan Oblast – Dobrovolnoye (now licensed).
- Khabarovsk Krai – Lastochka.
- Republic of Tyva – Ust-Uyuk and Onkazhinskoye.
- Republic of Khakassia – Primorskoye.

All together these projects have 76,600 tonnes of reasonably assured and inferred resources, plus 106,000 tonnes of less-certain 'undiscovered' resources.

Rosnedra published a list of deposits in the **Republic of Karelia, Irkutsk Region and the Leningrad Region** to be offered for tender in 2009. In particular, Tyumenskiy in Mamsko-Chuiskiy District of Irkutsk Region was to be offered for development, followed by Shotkusskaya ploshchad in Lodeinopolsky District of Leningrad Region. In Karelia Salminskaya ploshchad in Pitkyaranskiy District and the Karku deposit were offered. None of these 2009 offerings had reasonably assured or inferred resources quoted, only 'undiscovered' resources in Russia's P1 to P3 categories and it appears that none were taken up. In 2016 the Karelia Ministry of Natural Resources and Ecology acknowledged only one uranium deposit “of no commercial interest” at Srednyaya Padma (Medvezhegorsk District) and announced that no mining was planned.

Foreign and private equity in uranium mining

In October 2006 Japan's Mitsui & Co with Tenex agreed to undertake a feasibility study for a uranium mine in eastern Russia to supply Japan. First production from the Yuzhnaya mine in Sakha Republic (Yakutia) is envisaged for 2009. Mitsui had an option to take 25% of the project, and was funding $6 million of the feasibility study. Construction of the Yuzhnaya mine was estimated

to cost US$ 245 million, with production reaching 1000 tU/yr by 2015. This would represent the first foreign ownership of a Russian uranium mine. However, according to the 2016 *Red Book*, Yuzhnaya now appears to be part of the Elkon project (see above).

Following from previous deals with Tenex, in November 2007 Cameco signed an agreement with ARMZ. The two companies are to create joint ventures to explore for and mine uranium in both Russia and Canada, starting with identified deposits in northwestern Russia and the Canadian provinces of Saskatchewan and Nunavut.

In addition to ARMZ, private companies may also participate in tenders for mining the smaller and remote uranium deposits being prepared for licensing in Russia. ARMZ is open to relevant investment projects with strategic partners, and Lunnoye deposit is an example where a private company Zoloto Seligdara is partnering with ARMZ.

Mine rehabilitation

Some RUR 340 million (US$10m) is being allocated in the federal budget to rehabilitate the former Almaz mine in Lermontov, Stavropol Territory, in particular Mine 1 on Beshtau Mountain and Mine 2 on Byk Mountain, as well as reclamation of the tailings dump and industrial site of the hydrometallurgical plant. The work will be undertaken by Rosatom organizations under Rostechnadzor. In 2008, rehabilitation of Lermontovsky tailings was included in a federal target program, and over RUR 360 million was allocated for the purpose.

Secondary supplies

Some uranium also comes from reprocessing used fuel from VVER-440, fast neutron and submarine reactors - some 2500 tonnes of uranium has so far been recycled into RBMK reactors.

Also arising from reprocessing used fuels, some 32 tonnes of reactor-grade plutonium has been accumulated for use in MOX. Added to this there is now 34 tonnes of weapons-grade plutonium from military stockpiles to be used in MOX fuel for BN-600 and BN-800 fast neutron reactors at Beloyarsk, supported by a $400 million payment from the USA. Some of this weapons plutonium may also be used in the MHR high-temperature gas-cooled reactor under development at Seversk, if this proceeds.

About 28% of the natural uranium feed sent to USEC in USA for enrichment, and contra to the LEU supplied from blended-down Russian military uranium, is being sent to Russia for domestic use. The value of this to mid 2009 was US$ 2.7 billion, according to Rosatom. See also Military Warheads as Source of Fuel paper.

Russia's uranium supply is expected to suffice for at least 80 years, or more if recycling is increased. However, from 2020 it is intended to make more use of fast neutron reactors.

Fuel Cycle Facilities: conversion & enrichment

Many of Russia's fuel cycle facilities were originally developed for military use and hence are located in former closed cities (names bracketed) in the country. In October 2015 the ministry of economic development moved to open four of these which host facilities managed by Rosatom: Novouralsk, Zelenogorsk, Seversk and Zarechny.

In 2009 the conversion and enrichment plants were taken over by the newly-established JSC Enrichment & Conversion Complex, and in 2010 this became part of TVEL, a subsidiary of Atomenergoprom.

Seversk in Western Siberia is a particular focus of new investment, with Rosatom planning to spend a total of RUR100 billion on JSC Siberian Chemical Combine (SCC, SGChE) over 2012-20 to develop its “scientific, technical and production potential in terms of nuclear technology.” SCC comprises several nuclear reactors and plants for conversion, enrichment, separation and reprocessing of uranium and separation of plutonium. In 2012 Rosatom announced that it was investing RUR 45.5 billion ($1.6 billion) in SCC at Seversk to 2017 for modernising the enrichment capacity and setting up a new conversion plant.

TVEL has decided to rationalize some of its activities at Novouralsk, setting up a scientific and production association (SPA) in 2016 to incorporate Urals Gas Centrifuges Plant (UZGT or UGCP), Novouralsk Scientific and Design Center (NSDC), Uralpribor, and Electrochemical Converters Plant (ECCP).

Conversion

Russia's total uranium conversion capacity is about 25,000 tU/yr, but only about half of this is used as of 2013.

TVEL plans to consolidate its conversion capacity at JSC Siberian Chemical Combine (SCC) at Seversk near Tomsk, where some capacity already operates. In 2012 Rosatom said it would spend RUR 7.5 billion to set up a new conversion plant at SCC Seversk, to commence operation in 2016. The new plant is designed to have a capacity of 20,000 tU per year from 2020, including 2000 t of recycled uranium. Public hearings on the project were under way in 2014. The 2015 edition of the World Nuclear Association Nuclear Fuel Report gives capacity then as 12,500 tU.

The main operating conversion plant has been at Angarsk near Irkutsk in Siberia, with 18,700 tonnes U/yr capacity – part of TVEL's JSC Angarsk Electrolysis & Chemical Combine (AECC). In anticipation of the planned new plant at SCC Seversk however, the Angarsk conversion plant was shut down in April 2014.

TVEL also had conversion capacity at Kirovo-Chepetsky Chemical Combine (KCCC) in Glazoy, which was shut down in the 1990s. Since 2009 this has been a RosRAO site, for clean-up

The Elektrostal conversion plant, 50 km east of Moscow, has 700 tU/yr capacity for reprocessed uranium, initially that from VVER-440 fuel. It is owned by Maschinostroitelny Zavod (MSZ) whose Elemash fuel fabrication plant is there. Some conversion of Kazakh uranium has been undertaken for west European company Nukem, and all 960 tonnes of recycled uranium from Sellafield in UK, owned by German and Netherlands utilities, has been converted here. UK-owned recycled uranium has also been sent there.

Uranium enrichment

Four enrichment plants totalling 24 million kg SWU/yr of centrifuge capacity operate at Novo-Uralsk (formerly Sverdlovsk-44) near Yekaterinburg in the Urals, Zelenogorsk (formerly Krasnoyarsk-45), Seversk (formerly Tomsk-7) near Tomsk, and Angarsk near Irkutsk – the last three all in Siberia. The first two service foreign primary demand and Seversk specialises in enriching reprocessed uranium, including that from western Europe. As of early 2011, all are managed by TVEL, rather than Tenex (Techsnabexport).

Plant	Operator	Capacity (M SWU/yr)	Special features
Novouralsk	JSC Urals Electrochemical Combine	10	Can enrich to 30%
Zelenogorsk	PA ElectroChemical Plant (ECP)	8.7 (expanding to 12)	
Seversk	JSC Siberian Chemical Combine (SCC, SGChE)	3.0	RepU enrichment
Angasrk	JSC Angarsk Electrolysis & Chemical Combine	2.6	Tails enrichment
Total		24.3 (expanding to 27.6)	

The **Novouralsk** (Novo-Uralsk) plant is part of the JSC Urals Electrochemical Combine (UECC) in the Sverdlovsk region. It has operated 8th generation centrifuges since 2003, and 9th generation units from 2013. The fourth cascade of 9th generation centrifuges was commissioned in August 2016. TVEL is spending RUR 42 billion on re-equipping the plant with 9th generation units by 2019. In 2016 it was operating 6th to 9th generation centrifuges. The plant can enrich to 30% U-235 (for research and BN fast reactors), the others only to 5% U-235.

The TVEL-Kazakh JV Uranium Enrichment Centre (UEC) bought a 25% share of UECC and became entitled to half its output – up to 5 million SWU/yr (see below). In April 2013 the government commission for control over foreign investments approved this sale.

UECC once claimed 48% of Russian enrichment capacity and 20% of the world's. Rosatom in 2015 applied to the government to create a territory of priority development (TPD) in Novouralsk, a special economic zone enjoying low taxes, simplified administrative procedures and other benefits.

The **Zelenogorsk** plant is known as the PA Electrochemical Plant (ECP) in the Krasnoyarsk region (120 km east of that city), and has ISO 14001 environmental accreditation and ISO 9001 quality assurance system. It is starting to run 9th generation centrifuges and in 2021 commissioned its third cascade of these. In 2011 Rosatom said the plant's capacity was 8.7 million SWU/yr and it planned to increase that to 12 million SWU/yr by 2020, with a view to exporting its services. Rosatom was investing RUR 70 billion ($2.3 billion) by 2020 in developing the plant, with up to 90% of the

new centrifuges installed there to make it the main enrichment plant. It is the site of a new deconversion plant (see below).

The **Seversk** plant is part of the JSC Siberian Chemical Combine (Sibirsky Khimichesky Kombinat – SKhK or SCC), Tomsk region, which opened in 1953. It is about 15 km from Tomsk. As well as the enrichment plant with substantial capacity for recycled uranium the site has other facilities, and several plutonium production reactors (now closed). It is starting to run 9th generations centrifuges.

Angarsk, near Irkutsk in Siberia, is part of the JSC Angarsk Electrolysis & Chemical Combine (AECC). It is the only enrichment plant located outside a 'closed' city, nor has it had any defence role, and hence it became the site of the new International Uranium Enrichment Centre (IUEC) and fuel bank. In 2014 AECC said it would retain its present capacity. In December 2014 it started to undertake enrichment of tails (depleted UF_6) stored onsite up to natural UF_6 levels, and expects this to continue to 2030 as a major activity.

Technology: Diffusion technology was phased out by 1992 and all plants now operate modern gas centrifuges, with fitting of 8th generation equipment now complete. New units have a service life of up to 30 years, compared with half that previously. The last 6th & 7th generation centrifuges were set up in 2005, 8th generation equipment was supplied over 2004 to 2012, and about 240,000 units per year replaced 5th generation models. (6th generation units are still produced for export to China.) Two new 9th generation cascades were commissioned in 2015 and 10th generation units were being tested in 2016.

While TVEL had taken over responsibility for manufacture, in 2016 Rosatom decided to combine the design and production of centrifuges at the Urals Gas Centrifuge Plant (UZGT or UGCP) in Novouralsk, as part of the scientific and production association (SPA) set up by TVEL. OKB-Nizhniy Novgorod and Cetrotech-SPb had been involved in design and manufacture. The first 9th generation centrifuges were supplied to UECC early in 2013 from UZGT.

Tails re-enrichment: A significant proportion of the capacity of Novouralsk and Zelenogorsk plants – some 7 M SWU/yr – was earlier taken up by enrichment of tails (depleted uranium), including for west European companies Areva and Urenco. According to WNA sources, about 10,000 to 15,000 tonnes of tails per year, with U-235 assays between 0.25% and 0.40%, has been shipped to Russia for re-enrichment to about 0.7% U-235 since 1997. The tails

were stripped down to about 0.10% U-235, and remain in Russia, being considered a resource for future fast reactors. The contracts for this work for Urenco and Areva ended in 2010.

A portion of the Zelenogorsk capacity, about 4.75 M SWU/yr, was taken up with re-enrichment of tails to provide 1.5% enriched material for downblending much of the Russian HEU destined for USA. It was also the site for downblending much of the of ex-weapons uranium for sale to the USA (though all the other three plants may have contributed over the 20 years).

Seversk capacity is about 3 M SWU/yr, and some recycled uranium (from reprocessing) has been enriched here for Areva, under a 1991 ten-year contract covering about 500 tonnes UF_6. (French media reports in 2009 alleging that waste from French nuclear power plants was stored at Seversk probably refer to tails from enrichment of the recycled uranium.) It is understood to be enriching the 960 tU of reprocessed uranium from Sellafield in UK, belonging to its customers in Germany and Netherlands, sent to Elektrostal in eight shipments over 2001-09.

In 2012 Rosatom announced that it was investing RUR 45.5 billion ($1.6 billion) in SCC at Seversk to 2017 for modernising the enrichment capacity and setting up a new conversion plant.

Angarsk (AECC) is the smallest of three Siberian plants, with capacity of about 2.6 million SWU/yr. In July 2011 TVEL confirmed that there were no plans to expand it. A significant focus is tails enrichment. The International Uranium Enrichment Centre (IUEC) has been set up at Angarsk (see following IUEC section).

TVEL-Kazakh JV Uranium Enrichment Centre (UEC)

In the context of a December 2006 agreement with Kazakhstan, in 2008 Kazatomprom set up a 50-50 joint venture with Techsnabexport (Tenex) for financing a 5 million SWU/yr increment to the Angarsk plant, with each party to contribute about US$ 1.6 billion and hold 50% equity. It then appeared that initial JV capacity would be about 3 million SWU/yr, with first production in 2011. However, in 2010 Rosatom announced that this would not proceed, due to surplus world capacity, but other joint venture enrichment arrangements with Kazatomprom were offered, notably up to a 49% share in Novouralsk or Zelenogorsk.

After deciding that it would be uneconomic to expand capacity at Angarsk, in March 2011 it was announced that Kazatomprom would buy a share in Urals Electrochemical Combine (UECC) which owns the Novouralsk plant through its 50% equity in the TVEL-

Kazakh JV Uranium Enrichment Centre (UEC), "instead of building new capacity at AECC" at Angarsk where UEC was originally established. In mid-2011 it was reported that Kazatomprom would acquire shares in UECC either directly (30%) or in the event as a 50% shareholder in UEC with TVEL, related to the need to enrich 6000 tU/yr. Over 2012-13 UEC acquired 25% of UECC, and UEC became operational in the second half of 2013, with access to 5 million SWU/yr – about half of UECC production. The cost of the Kazatomprom share, earlier estimated by it at $500 million, was not disclosed. The first batch of enriched uranium was shipped in November 2013. UEC share of production in 2014 was 4.99 million SWU.

Deconversion

Russia's W-ECP or W-EKhZ deconversion plant is at Zelenogorsk Electrochemical Plant (ECP). The 10,000 t/yr deconversion (defluorination) plant was built by Tenex under a technology transfer agreement with Areva NC (now Orano), so that depleted uranium can be stored long-term as uranium oxide, and hydrogen fluoride is produced as a by-product. The W1-ECP plant is similar to Areva's W2 plant at Pierrelatte in France and has mainly west European equipment. It was commissioned in December 2009 and to January 2021 had processed 100,000 t depleted uranium hexafluoride. The Russian-designed phase 2 for production of anhydrous hydrogen fluoride was commissioned in December 2010. During the ten years to end of 2020, some 11,000 t of anhydrous hydrogen fluoride as well as much more hydrofluoric acid were shipped to customers. TVEL is building a second unit, W2-ECP, with equipment from Orano Projects in France. This will expand ECP's capacity to 20,000 t/yr depleted uranium hexafluoride from 2023 and producing up to 2400 t/yr of anhydrous hydrogen fluoride.

Fuel fabrication

Fuel fabrication is undertaken by JSC TVEL, which supplies 76 nuclear reactors in Russia and 13 in other countries as well as 30 research reactors and fuel for naval and icebreaker reactors. Its operations are certified against ISO 9001 and it has about 17% of the world market for fabricated fuel. Russian fuel technology is supported by TVEL's A.A. Bochvar High Technology Research Institute of Inorganic Materials (**VNIINM**).

Fuel cycles

Russia aims to maximise recycling of fissile materials from used fuel. Hence reprocessing used fuel is a basic practice, with reprocessed uranium being recycled and plutonium used in MOX, at present only for fast reactors. However, innovative developments of MOX use open up wider possibilities, and both the REMIX cycle and the Dual Component Power System are described below.

Uranium fuel fabrication

TVEL has two fuel fabrication plants with combined capacity of 2800 t/yr finished fuel:

- The huge Maschinostroitelny Zavod (MSZ) at Elektrostal 50 km east of Moscow – known as Elemash.
- Novosibirsk Chemical Concentrates Plant (NCCP) in Siberia.

TVEL's Chepetsk Mechanical Plant (CMP or ChMZ) near Glazov in Udmurtiya makes zirconium cladding and also some uranium products.

Most fuel pellets for RBMK and VVER-1000 reactors were being made at the Ulba plant at Ust Kamenogorsk in Kazakhstan, but Elemash and Novosibirsk have increased production. MSZ/Elemash produces fuel assemblies for both Russian and west European reactors using fresh and recycled uranium. It also fabricates research reactor and icebreaker fuel and in 2016 is producing the first fuel for the RITM-200 reactors in new icebreakers. VNIINM claims the fuel has greater energy density than previous icebreaker fuel.

Novosibirsk produces mainly VVER-440 & 1000 fuel, including that for initial use in China.

MSZ/Elemash is the principal exporter of fuel assemblies. Total production is about 1400 t/yr, including fuel assemblies for VVER-440, VVER-1000, RBMK-1000, BN-600 reactors, powders and fuel pellets for delivery to foreign clients. It has a contract to supply high-enriched uranium (HEU) fuel over seven years for China's first CFR600 fast reactor. The plant also produces nuclear fuel for research reactors.

TVEL is developing a uranium-erbium fuel for VVERs enriched to 5-7% for load-following and longer fuel cycles. Some RBMK fuel is already enriched over 5%.

Early in 2021 MSZ set up a new production line for fast reactor fuel, including HEU. Russia's BN-600 reactor uses uranium fuel with three levels of enrichment: 17%, 21% and 26%. Fuel for China's CFR600 is likely to be similar. On another production line MSZ has already provided fuel for China's CEFR, including a 2020 reload, reported to be 64% enriched.

TVEL's NCCP also produces pure lithium-7, and accounts for over 70% of the world supply of Li-7, both 99.95% for use in PWR cooling systems, and also now 99.99% pure. A plant upgrade in 2013 makes it possible to double the volume of Li-7 output there.

TVEL has done extensive work done on utilization of reprocessed uranium (RepU) in VVER-type reactors, and there are plans for all units of the Kola nuclear station to shift to RepU fuel. Some PWR reactors, e.g. Kalinin 2 and Balakovo 3, are using recycled uranium in TVSA fuel assemblies already.

There is no plan or provision to use MOX in light-water reactors.

TVEL owns 35% equity in the Ulba Metallurgical Plant in Kazakhstan. This has major new investment under way. It has secured both ISO 9001 and ISO 14001 accreditation. Since 1973 Ulba has produced nuclear fuel pellets from Russian-enriched uranium which are used in Russian and Ukrainian VVER and RBMK reactors. Some of this product incorporates gadolinium and erbium burnable poisons. Ulba briefly produced fuel for submarines (from 1968) and satellite reactors. Since 1985 it has been able to handle reprocessed uranium, and it has been making fuel pellets incorporating this for western reactors, supplied through TVEL.

TVEL's Moscow Composite Metal Plant designs and makes control and protection systems for nuclear power reactors.

REMIX fuel cycle

REMIX (Regenerated Mixture) fuel has been developed by the V.G. Khlopin Radium Institute for Tenex as a development of MOX to supply light water reactors. Remix fuel is produced directly from a non-separated mix of recycled uranium and plutonium from reprocessing used fuel, with a low-enriched uraniium (up to 17% U-235) make-up comprising about 20% of the mix. This gives fuel with about 1% Pu-239 and 4% U-235 which can sustain burn-up

of 50 GWd/t over four years and has similar characteristics to normal LWR fuel. It is distinct from MOX in having low and incidental levels of plutonium – none is added. The spent Remix fuel after four years is about 2% Pu-239* and 1% U-235, and following about five years of cooling and then reprocessing the non-separated uranium and plutonium is recycled again after LEU addition. The waste (fission products and minor actinides) is vitrified, as today from reprocessing, and stored for geological disposal. Before vitrification it may be processed to recover valuable fission products such as isotopes Cs, Sr and Tc.

* a 68% increase, compared with 104% in MOX fuel cycle, according to Tenex.

Remix fuel can be repeatedly recycled with 100% core load in current VVER-1000 reactors and correspondingly reprocessed many times – up to five times, so that with fewer than three fuel loads in circulation a reactor could run for 60 years using the same fuel, with LEU recharge. As with normal MOX, the use of Remix fuel reduces consumption of natural uranium in VVERs by about 20% at each recycle as compared with open fuel cycle. Remix can serve as a replacement for existing reactor fuel, but in contrast to MOX there is a higher cost for fuel fabrication due to the high activity levels from U-232. Compared with UO_2 fuel, the cost increment is 25-30%. The Remix cycle can be modified from the above figures according to need. The increasing concentrations of even isotopes of both elements is compensated by the fresh uranium top-up, possibly at increasing enrichment levels.

A 2019 study showed that the use of regenerated uranium in Remix fuel for VVER reactors, and therefore the U-236 isotope, also significantly increases the proportion of Pu-238 in the fuel, which prevents its diversion for non-peaceful purposes.

Remix allows all the recovered uranium and plutonium to be recycled and will give a saving in used fuel storage and disposal costs compared with the once-through fuel cycle, matched by the reprocessing cost, though this is expected to reduce. Compared with the MOX cycle, it has the virtue of not giving rise to any accumulation of reprocessed uranium (RepU) or allow any separated plutonium.

Rosatom loaded three TVS-2M fuel assemblies each with six REMIX fuel rods into Balakovo 3 in June 2016. They remained for two fuel cycles, and a third 18-month cycle began in early 2020. These all showed good results, and Rosatom is now proceeding to pilot operation of several full-REMIX fuel assemblies. No changes in

reactor design or safety measures are required. Remix fuel is also being tested in the MIR research reactor at RIAR in Dimitrovgrad.

Tenex suggests Remix being used with a form of fuel leasing from a supplier to a utility, with repeated recycle between them. Commercial application is planned for the mid-2020s.

In August 2020 Rosatom announced that Remix fuel for VVER-1000 reactors would be produced on a new production line at the Siberian Chemical Plant (SCC) at Seversk from 2023. In June 2021 TVEL commissioned equipment for the pilot fuel production line, enabling initial production of fuel assemblies by year end, using fuel pellets made at the MCC Zheleznogorsk plant. Eventually a commercial-scale Remix fuel fabrication plant is envisaged.

MOX fuel fabrication (only for fast reactors)

In late 2007 it was decided that MOX fuel production using recycled materials should be based on electrometallurgical (pyrochemical) reprocessing and vibropack dry processes for fuel fabrication, as developed at RIAR. The goals for closing the fuel cycle included minimising cost, recycle of minor actinides (for burning), excluding separated plutonium, and arrangement of all procedures in remote systems to allow for 'hot' materials. However, plans for vibropack fuels are not being pursued with any vigour.

MCC Zheleznogorsk MOX plant: A 60 t/yr commercial mixed oxide (MOX) fuel fabrication facility (MFFF) commenced operation at Zheleznogorsk (formerly Krasnoyarsk-26, 70 km northeast of Krasnoyarsk) in 2015, operated by the Mining & Chemical Combine (MCC or GKhK). This was built at a cost of some RUR 9.6 billion as part of Rosatom's Proryv, or 'Breakthrough', project, to develop fast reactors with a closed fuel cycle whose MOX fuel will be reprocessed and recycled. It represents the first industrial-scale use of plutonium in the Russian civil fuel cycle, and is also the Russian counterpart to the US MFFF for disposition of 34 tonnes of weapons-grade plutonium.* About half the plant's equipment was imported.

* The head of Rosatom reported to the president in September 2015: "Industrial operation has begun at a new MOX fuel (uranium-plutonium fuel) production plant, the first such plant in history. Our American partners have still not managed to finish the plant they were building. They have already spent $7.7 billion on it and, as Congress informs, they are now going to suspend the project because no one knows how much more money it will cost. We built

our plant in 2.5 years at a cost of a little over $200 million, or 9.6 billion rubles. The plant is working and is now reaching industrial capacity."

MCC's MFFF will make 400 pelletised MOX fuel assemblies per year for the BN-800 and future BN-1200 fast reactors. The MOX can have up to 30% plutonium. The capacity is designed to be able to supply five BN-800 units or equivalent BN-1200 capacity. First production of 20 fuel assemblies for Beloyarsk 4 was in 2015, working up to full capacity in 2017. The BN-800 each year requires 1.84 tonnes of reactor-grade plutonium recovered from 190 tonnes of used VVER fuel. The first serial batch of MOX for BN-800 passed acceptance tests in December 2018. (Plutonium from used BN fuel will be used in VVER-1000 reactors.) The MFFF is built in rock tunnels at a depth of about 200 metres.

Longer-term MCC Zheleznogorsk was intending to produce MOX granules for vibropacked fuel using civil plutonium oxide, ex-weapons plutonium metal and depleted uranium. Initial capacity of 14 t/yr of granules was funded to RUR 5.1 billion (US$ 169 million then) over 2010-12. The granulated MOX is sent to RIAR Dimitrovgrad for vibropacking into FNR fuel assemblies.

In June 2011 Rosatom announced that it was investing RUR 35 billion in MCC to 2030, including particularly MOX fuel fabrication. In February 2012 the figure was put at RUR 80 billion minimum.

Mayak MOX plant: A small pelletised MOX fuel fabrication plant has operated at the Mayak plant at Ozersk since 1993, for BN-350 and BN-600 fuel (40 fuel assemblies per year), and it supplied some initial pelletised MOX fuel for BN-800 start-up, the assemblies being made by RIAR Dimitrovgrad.

Seversk MOX plant: Another MOX plant for disposing of military plutonium is planned at Seversk (Tomsk-7) in Siberia, to the same design as its US equivalent. This is for dense MOX fuel for fast reactors, and was planned for completion by the end of 2017, with RUR 5.8 billion allocated by TVEL for the equipment. (Seversk had the other two dual-purpose but basically military plutonium production reactors, totalling 2500 MWt. One of these – ADE4 – was shut down in April 2008, the other – ADE5 – in June 2008.)

RIAR Dimitrovgrad MOX plant: The Research Institute of Atomic Reactors (RIAR or NIIAR) at Dimitrovgrad, Ulyanovsk, has a small MOX fuel fabrication plant. This produces vibropacked fuel which was said to be more readily recycled. Under the federal target programme this was allocated RUR 2.95 billion (US$ 83 million) for expansion from 2012. Its main research has been on the use of

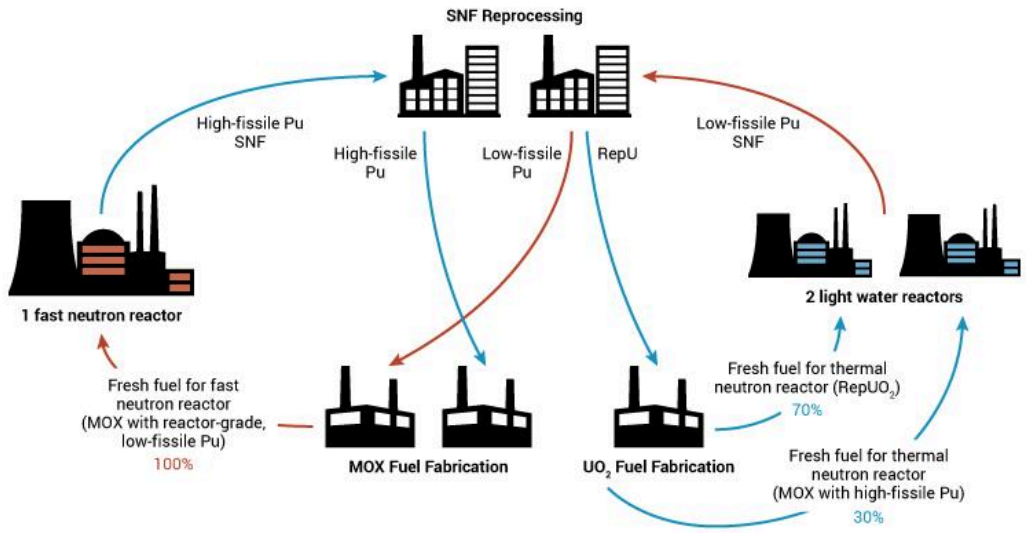
military plutonium in MOX, in collaboration with France, USA and Japan. From 2014 the plant produced 106 fuel assemblies for Beloyarsk 4 BN-800, before MCC's MFFF took over this role.

Vibropacked MOX fuel (VMOX) was earlier seen as the way forward. This is made by agitating a mechanical mixture of $(U,Pu)O_2$ granulate and uranium powder, which binds up excess oxygen and some other gases (that is, operates as a getter) and is added to the fuel mixture in proportion during agitation. The getter resolves problems arising from fuel-cladding chemical interactions. The granules are crushed $(U,Pu)O_2$ cathode deposits from pyroprocessing. VMOX needs to be made in hot cells. It has been used in BOR-60 since 1981 (with 20-28% Pu), and tested in BN-350 and BN-600 as part of a hybrid core (with some military plutonium). This was evaluated by OKBM and Japan Nuclear Cycle Development Institute. However, its future is uncertain, and MOX fuel may revert to being conventional sintered pellets.

Dual-component power system MOX

Rosatom has proposed a fuel cycle involving both thermal and fast reactors, using two kinds of MOX fuel, and envisages implementing this system when the first BN-1200 reactors are online about 2027. In 2020 the first MOX using plutonium from conventional power reactors was loaded into Beloyarsk's BN-800 reactor and later in the year another 180 such fuel assemblies will be added. By the end of 2021, the reactor will fully switch to MOX fuel.

Balanced Arrangement for Dual-Component Nuclear Power System



Source: Rosatom

In this fuel cycle, normal thermal reactors are the primary plutonium source, but this plutonium is reactor-grade, with about one-third

even-mass number non-fissile isotopes. The plutonium is mixed with deflourinated tails from uranium enrichment (*i.e.* depleted uranium). Whether derived from used uranium fuel or MOX fuel, it is separated and made into MOX fuel for fast breeder reactors with not less than 1.2 breeding ratio, and the used fuel from these has a much lower proportion of even-number non-fissile plutonium isotopes.

In future this 'clean' or high-fissile plutonium recovered from fast reactor fuel can then made into MOX fuel for the original thermal reactors, and comprise about 30% of their fuel. The other 70% could be enriched reprocessed uranium (RepU), the depleted tails of which are also used for MOX, instead of using normal depleted uranium. Their used fuel is reprocessed to continue the dual cycle. Minor actinides are burned in the fast reactors.

One fast reactor running on 'dirty' MOX would therefore be in balance with two VVER reactors fuelled with 'clean' MOX (30% of load) and RepU oxide enriched to about 17% U-235 (70% of load) via segregated reprocessing facilities and segregated fuel fabrication.

Further details are in the information paper on Mixed Oxide Fuel.

Nitride fuel fabrication for fast reactors

Overall, RUR 17 billion is budgeted for nitride fuel development, which is mainly for the BREST-300 reactor, part of Rosatom's *Proryv* or 'Breakthrough' project. Both SCC plants will be part of the Pilot Demonstration Power/Energy Complex (PDPC or PDEC) with the BREST reactor, integral to the *Proryv* project and approved by government decree in August 2016. The *Proryv* project at SCC is expected to be fully operational from 2023.

To avoid problems in reactor operation and spent fuel, nitrogen-15 is the preferred isotope. VNIINM has patented a technique for enrichment in N-15, annual demand for which is expected to be several tonnes.

SCC nitride fuel plant KEU-1: In collaboration with TVEL, the Siberian Chemical Combine (SCC) at Seversk is making test batches of dense mixed nitride uranium-plutonium (MNUP) fuel for fast reactors, essentially prototype fuel for BREST. Construction of SCC's pilot nitride fuel plant started in March 2014 with a view to commissioning in 2017-18, in time to produce fuel for the first BREST-300 reactor, which is now expected in operation about

2024. In April 2016 Atomenergomash supplied to SCC a plant for preparation of input materials for automated fabrication of MNUP fuel for fast neutron reactors.

SCC completed acceptance tests on the first ETVS nitride fuel assembly in September 2014, and it had further ones (ETVS-10 & 11) ready a year later, using parts supplied by VNIINM. In April 2015 the first ETVS nitride fuel assemblies were put into the BN-600 reactor at Beloyarsk for testing over three years, and by August 2015 there were nine ETVS there. In November 2015 the post-irradiation inspection of ETVS-1 after six-month storage to cool showed it to be in good shape. In April 2016 two more dense nitride fuel assemblies (ETVS-12 & 13) were delivered to Beloyarsk for irradiation in the BN-600 reactor. They were designed by VNIINM and made by SCC as prototypes for BREST-300 and BN-1200 reactors. In mid-2016 VNIINM produced two more pilot fuel assemblies, ETVS-14 & 15, with mixed nitride fuel for testing in the BN-600 reactor at Beloyarsk. MSZ completed acceptance tests on these in August. In December 2016 SCC announced successful post-irradiation tests on ETVS fuel assemblies, confirming their suitability for BREST. ETVS-16 to 21 were scheduled for 2017. The next series of ETVS will be of a different design. By November 2020, more than 1000 MNUP fuel rods had been produced and more than 21 fuel assemblies had been irradiated in BN-600, the latest ones each with 61 fuel rods.

SCC nitride fuel plant KEU-2: SCC started construction of a second integrated experimental facility (KEU-2) in 2016, to fabricate fuel for testing in the BN-800 reactor at Beloyarsk. A U-Pu-Np nitride fuel fabrication and recycling facility is part of the Pilot Demonstration Power Complex (PDPC; Russian acronym: ODEK) at SCC. Rosatom began installing equipment here for MNUP fuel fabrication and refabrication for the BREST-300 in 2017. The main fabrication line was expected in operation in 2020, with daily production capacity of up to 60 kg of fuel, or 120 nuclear fuel assemblies, and a total of 14.7 tonnes of fuel per year.

In October 2014 SCC announced a tender for a reprocessing plant to be completed by 2018, with VNIPIET as SCC's preferred bidder. It included a module for processing used nuclear fuel, to examine technologies VNIINM and the VG Khlopin Radium Institute have developed. VNIINM said its experiments in 2016 had confirmed for the first time that the technology used for the reprocessing of used mixed nitride fuel enables the re-use of more than 99.9% of the actinides. The actual RUR 20 billion plant is to have a capacity of 5 t/yr used fuel from the BREST-300 and 0.5 t/yr of "rejects from

electrolysis process and americium-containing burning elements." It will commence operation about 2024, after the BREST-300 is in service. This will be part of the Pilot Demonstration Power/Energy Complex (PDPC or PDEC) with the BREST reactor.

SCC started testing three different refining technologies for the plant in 2016. The best option will be selected and used in the used fuel recycling module within PDPC. The project manager said that the refining installation "can be used as a sector-wide test-bench to deal with uranium, plutonium, and neptunium."

Mayak nitride fuel plant: A new 14 tonne per year plant to fabricate dense mixed nitride fuel for fast neutron reactors is planned at PA Mayak, to operate from 2018. In the federal target programme to 2020, RUR 9.35 billion ($310 million) was budgeted for it. Later it may be expanded to 40 tonnes per year.

International Uranium Enrichment Centre (IUEC)

The IUEC concept was inaugurated at the end of 2006 in collaboration with Kazakhstan, and in March 2007 the International Atomic Energy Agency (IAEA) agreed to set up a working group and continue developing the proposal. In September 2007 the joint stock company Angarsk International Uranium Enrichment Centre (JSC Angarsk IUEC) was registered and a year later Rostechnadzor licensed the centre.

Late in 2008 Ukraine's Nuclear Fuel Holding Company, SC Nuclear Fuel, decided to take a 10% stake in it, matching Kazatomprom's 10%, and this was effected in October 2010. Armenia finalised its 10% share in IUEC in May 2012 (2600 shares for RUR 2.6 million). Negotiations since then have proceeded with South Africa, Vietnam, Bulgaria, UAE, Jordan, South Korea and Mongolia (in connection with Russian uranium interests there). Russia also invited India to participate in order to secure fuel for its Kudankulam plant. The aim is for Techsnabexport/TVEL eventually to hold only 51%. Each of the 26,000 IUEC shares is priced at RUR 1000.

Present equity in JSC Angarsk IUEC: TVEL 70%, Kazatomprom 10%, Ukraine State Concern Nuclear Fuel 10%, Armenia NPP 10%.

The centre is to provide assured supplies of low-enriched uranium for power reactors to new nuclear power states and those with small nuclear programmes, giving them equity in the project, but without allowing them access to the enrichment technology. Russia will maintain majority ownership. IUEC will sell both enrichment

services (SWU) and enriched uranium product. Arrangements for IAEA involvement were being sorted out in 2009, and in 2010 a feasibility study commenced on IUEC investment, initially for equity in JSC Angarsk Electrolysis & Chemical Combine (AECC) so that part of its capacity supplies product to IUEC shareholders.

The existing enrichment plant at **Angarsk** was to feed the IUEC and accordingly was removed from the category of "national strategic installations", though it had never been part of the military programme. In February 2007 the IUEC was entered into the list of Russian nuclear facilities eligible for implementation of IAEA safeguards. The USA has expressed support for the IUEC at Angarsk. Since 2010 the facility has been under IAEA safeguards.

Development of the IUEC was envisaged in three phases:

1. Use part of the existing capacity at Angarsk in cooperation with Kazatomprom and under IAEA supervision.
2. Expand Angarsk capacity (perhaps double) with funding from new partners by 2017.
3. Full internationalisation with involvement of many customer nations under IAEA auspices.

In 2012-13 the IUEC website said: “The JSC IUEC has been established within the Angarsk Electrolysis Chemical Complex, but it can use capacities of other three Russian combines to diversify production and optimize logistics.”

In 2016 a major customer was Ukraine’s State Concern Nuclear Fuel, which since 2012 has bought 60,000 SWU per year, proportional to its shareholding.

IUEC guaranteed LEU reserve ('fuel bank')

In November 2009 the IAEA board approved a Russian proposal to create an international **guaranteed LEU reserve** or 'fuel bank' of low-enriched uranium under IAEA control at the IUEC at Angarsk. This was established a year later and comprises 123 tonnes of low-enriched uranium as UF_6, enriched 2.0-4.95% U-235 (with 40t of latter), available to any IAEA member state in good standing which is unable to procure fuel for political reasons. It is fully funded by Russia, held under safeguards, and the fuel will be made available to the IAEA at market rates, using a formula based on spot prices. Following an IAEA decision to allocate some of it, Rosatom will transport material to St Petersburg and transfer title to the IAEA, which will then transfer ownership to the recipient. The 120 tonnes of low-enriched uranium as UF_6 is equivalent to two full fuel loads

for a typical 1000 MWe reactor, and in 2010 was worth some $250 million.

This initiative complements the IAEA LEU Bank set up in Kazakhstan by making more material available to the IAEA for assurance of fuel supply to countries without their own fuel cycle facilities. The IAEA LEU Bank is located at the Ulba Metallurgical Plant (UMP) in Kazakhstan, which has 50 years of experience in handling UF_6. A formal agreement with Kazakhstan to establish the legal framework was signed in August 2015, and the partnership agreement between the IAEA and UMP was signed in May 2016. Construction of the building with 600 m^2 storage area started in September 2016, and the facility was formally opened at the end of August 2017. It became operational in 2019, and it awarded contracts to Orano and Kazatomprom to supply it.

Used fuel and reprocessing

Russian policy is to close the fuel cycle as far as possible and utilise recycled uranium, and also to use plutonium in MOX fuel. However, its achievements in doing this have been limited – in 2011 only about 16% of used fuel was reprocessed, this being from VVER–440s, BN-600, research reactors and naval reactors. The reprocessed uranium (RepU) is mainly used for RBMK fuel. By 2030 Rosatom hopes to fully close the fuel cycle. Commercial reprocessing started in 1977, and several projects at two sites have been under way to progress this intention:

- At Mayak Production Association in Ozersk, the RT-1 spent fuel reprocessing facility was first updated and returned to service in 2016, and will then be shut down in about 2030.
- At Mining and Chemical Combine (MCC) in Zheleznogorsk, the MOX fuel fabrication plant for fast reactors was commissioned in 2015 (see above).
- At MCC the Pilot Demonstration Centre (PDC) for used nuclear fuel reprocessing was commissioned in 2015.
- At MCC the full-scale RT-2 facility would be completed by 2025 to reprocess VVER, RBMK and BN used fuel into mixed-oxide (MOX) fuel or into REMIX – the regenerated mixture of uranium and plutonium oxides.

- At MCC Zheleznogorsk the spent fuel pool storage would be supplemented by dry storage, commissioned in 2012, and MCC will become the destination for all of Russia's used fuel.

In 2013 used fuel arisings in Russia were:

VVER-440	6 units	87 t/yr
VVER-1000	11 units	230 t/yr
RBMK	11 units	550 t/yr
EPG-6	4 units	4 t/yr
BN-600	1	6 t/yr
Naval, research	80+	?
Total		880 t/yr

All used fuel is stored at reactor sites for at least three years to allow decay of heat and radioactivity. High burn-up fuel requires longer before it is ready to transport. At present the used fuel from RBMK reactors and from VVER-1000 reactors is stored (mostly at reactor sites) and not reprocessed. It is expected that used fuel in storage will build up to about 40,000 tonnes by the time substantial reprocessing at MCC Zheleznogorsk gets under way about 2022. The materials from this will be burned largely in fast reactors by 2050, when none should remain.

In late 2007 it was decided that MOX fuel production using recycled materials from both light water and fast reactors should be based on electrometallurgical (pyrochemical) reprocessing. The goals for closing the fuel cycle are minimising cost, minimising waste volume, recycle of minor actinides (for burning), excluding separated plutonium, and arrangement of all procedures in remote-handled systems. This reprocessing route remains to be developed.

In August 2016 a new program for management of used fuel to 2020 was announced. It provides for transport of used fuel to Mayak at Ozersk for reprocessing, or to a central storage facility at MCC Zheleznogorsk where the reprocessing plant is due to be commissioned.

RT-1 reprocessing plant, Mayak

Used fuel from VVER-440 reactors Kola 1-4 and Rovno 1-2 in Ukraine), the BN-600 (Beloyarsk) and from naval reactors is sent to the Mayak Chemical Combine's 400 t/yr RT-1 plant (Chelyabinsk-65) at Ozersk, near Kyshtym 70 km northwest of Chelyabinsk in the Urals for reprocessing.* An upgrade of the RT-1 plant to enable it to take VVER-1000 fuel was completed in 2016, and reprocessing of fuel from Rostov began late in the year. In 2017, 20 tonnes of used VVER-1000 fuel from Balakovo is to be reprocessed.

* The original reprocessing plant at the site was hastily built in the mid-1940s, for military plutonium production in association with five producer reactors (the last shut down in 1990).

The RT-1 plant started up in 1971 and employs the Purex process. Since about 2000 the plant has been extended and modified so that it can accept a wide variety of inputs, including U-Be research reactor fuel. It had reprocessed about 5000 tonnes of used fuel to 2012 and was reported to be running at about 100 t/yr capacity, following the loss of foreign contracts. In 2015 RT-1 processed 230 tonnes of fuel, 35% more than in 2014, and its capacity is expected to reach 400 t/yr “within several years”, comprising all types from Russian designed reactors, notably VVER-1000 and RBMK. From 2017 it will also be able to reprocess uranium nitride fuel. However, after the commissioning of the RT-2 plant at MCC, it is due to be decommissioned about 2030.

About 93% of its feed to 2015 has been from Russian and Ukrainian VVER-440 reactors, about 3% from naval sources or icebreakers and 3% from the BN-600 reactor. It earlier reprocessed BN-350 used fuel. Damaged used fuel is to be reprocessed there to avoid the need for prolonged storage. In September 2015 Rosatom said that reprocessing the fuel from 201 decommissioned vessels transferred to it from the Ministry of Defence was 97% complete, and that no naval fuel remained in the Far East. Regular shipments of used submarine fuel from Andreeva Bay storage to Mayak for reprocessing commenced in mid-2017, and 22,000 naval fuel assemblies are expected to be shipped by 2024, via Murmansk.

In 2015 Mayak started reprocessing the uranium-beryllium fuel from dismantled *Alfa*-class submarines, as a ‘nuclear legacy project’. These unsuccessful vessels had a single reactor of 155 MWt cooled by lead-bismuth and using very highly enriched uranium – 90% enriched U-Be fuel. The experience gained with lead-bismuth eutectic is being applied in Russia’s fast reactor programme – notably BREST (since SVBR was dropped).

Recycled uranium is enriched to 2.6% U-235 by mixing RepU product from different sources and is used in all fresh RBMK fuel, while separated plutonium oxide is stored. High-level waste is vitrified and stored. There are plans to use RepU for all the Kola VVER reactors. Vitrified HLW from Ukraine's VVER-440 used fuel is to be returned to Ukraine from 2018.

Used fuel storage capacity there is being increased from 6000 to 9000 tonnes, but will remain limited compared with Zheleznogorsk. Hence the used fuel received is usually treated fairly promptly. In 2015, 5184 RBMK used fuel assemblies were sent there from the Leningrad and Kursk plants, for storage initially.

Zheleznogorsk MCC: Pilot Demonstration Centre and RT-2 reprocessing plant

A **Pilot Demonstration Centre (PDC)** for several reprocessing technologies is operated by MCC at Zheleznogorsk, built at a cost of RUR 8.4 billion and completed in 2015 as a "strategic investment project". Its initial capacity with research hot cells is 10 t/yr, increasing to 100 t/yr, with later increase to 250 t/yr from 2018 as phase 2. PDC phase 2 was expected to be in full operation in 2019. It will have innovative technology including embrittlement by crystallization, and simultaneous gas, thermo and mechanical spent fuel assembly shredding. Initially it will deal with VVER-1000 fuel, later with fuel from fast reactors. It will effectively be the first stage of the large redesigned RT-2 plant at the MCC/GHK site to be operational about 2024. The cost of RepU product is expected to be some €500/kg. The PDC "can be used for demonstration of the closed nuclear fuel cycle of thermal neutron reactors running on REMIX-fuel" as well as producing MOX fuel.

The **RT-2 reprocessing plant** at Zheleznogorsk is now on track for completion with 700 t/yr capacity by 2025 (in addition to the 250 t/yr at PDC). Another 800 t/yr is planned by 2028. Originally it was planned to have two 1500 t/yr lines, but for some time the project was under review. Construction started in 1984 but halted in 1989 when 30-40% complete due to public opposition and lack of funds (though in 1993 it was officially reported as "under construction"). It has now been redesigned and is expected to operate from around 2025 with advanced Purex process, for both VVER-1000 and RBMK fuel, and also BN fuel. Its cost is about $2 billion, with no federal funds. The facility could form part of the new Global Nuclear Infrastructure Initiative and foreign equity in a joint stock company

is being considered. (See also International Collaboration section below.)

Zheleznogorsk MCC: RBMK and VVER used fuel storage

VVER-1000 used fuel is sent to the Mining & Chemical Combine (MCC) (Gorno-Khimichesky Kombinat – GHK) at Zheleznogorsk (Krasnoyarsk-26) in Siberia for pool storage. The site is about 60 km north of Krasnoyarsk. This fuel comes from three Russian, three Ukrainian and one Bulgarian plants. A large pool storage facility was built by MCC at Zheleznogorsk in 1985 for VVER-1000 used fuel, though its 6000 tonne capacity would have been filled in 2010. The facility was fully refurbished over 2009-10, and some dry storage capacity was commissioned in 2011. In December 2009 Rostechnadzor approved pool storage expansion to 7200 tonnes and MCC sought approval to expand it to 8400 tonnes capacity to allow another 6 years input. It is now planned to expand wet storage for VVER-1000 fuel to 11,000 tonnes.

In 2012 the first stage of an 8600 tonne **dry storage facility** for used fuel (INF DSF-2) was commissioned at Zheleznogorsk. It was built by the E4 Group at a cost of about $500 million for the MCC/GHK. It is the largest dry storage facility in the world, holding 8129 tonnes of RBMK fuel, initially from Leningrad and Kursk power plants, followed by Smolensk. At Leningrad the fuel is cut up and put into the large containers before being shipped to MCC. RBMK fuel is not presently economic to reprocess so has been stored at reactor sites, and when transferred to MCC it is stored in hermetically sealed capsules filled with nitrogen and helium, inside a building but air-cooled.

The second stage of MCC **dry storage** will take VVER-1000 fuel currently in wet storage there and increase capacity to over 37,000 tonnes (26,510 t RBMK, 11,275 t VVER). MCC expects to commission it about the end of 2016. It is expected to be commissioned about the end of 2015. The original wet storage facility is to be decommissioned in 2026. Used fuel will be stored for up to 50 years, pending reprocessing. MCC has flagged the possibility of storing foreign VVER-1000 used fuel, such as that from fuel take-back arrangements linked to foreign reactor sales (initially Iran). This can be reprocessed in Russia, but the waste must be repatriated.

Bilibino's LWGR used fuel is stored at Bilibino site.

(Three decommissioned graphite-moderated reactors which principally produced military plutonium, with associated underground reprocessing plant, are also at MCC Zheleznogorsk. The huge underground complex, 200-250 m deep, was originally established in 1950 for plutonium and weapons production.)

Other reprocessing plants

At SCC Seversk a reprocessing plant for nitride fuel from BREST fast reactors is envisaged to operate from 2024, closing that fuel cycle. See above under SCC nitride fuel plant KEU-2.

In 2016 it was announced that decommissioning of the HEU downblending and mixing plant at SCC would be completed by 2022. The plant was built in 1996 at the conversion plant in order to implement the Russia-US program for blending down high-enriched uranium from Russian nuclear weapons into low-enriched uranium for export and use in US nuclear power plants. This program concluded in 2013.

Some kind of radioactive waste processing plant is under construction at the Kursk nuclear power station, according to Nikimt-Atomstroy. A completed section, fully operational by the end of 2014, would process liquid radioactive waste. The two remaining sections of the project include a processing facility for solid radioactive waste and a storage facility.

Legacy materials

Russia has a significant amount of legacy materials, some as a result of military materials production (*e.g.* slightly irradiated uranium), others from the civil fuel cycle (*e.g.* reprocessed uranium), and as a result of reviews over 2006-08 these are now recognised as potentially having significant value. The total quantity is not such as to impact the civil market; there are some technical challenges (*e.g.* limiting U-232 to 5 ppb in enriched RepU), and in any case Russia's preference is to use the material domestically while making resultant expertise available internationally.

The main material not found in the civil nuclear fuel cycle is slightly irradiated uranium (SIU, 0.65% U-235) from military plutonium production with low burn-up of natural uranium, after reprocessing to separate that plutonium. If SIU is enriched, the product can readily be used in nuclear plants and the tails become DSIU, with

lower content of even uranium isotopes (232, 234, 236) than normal RepU, hence more valuable.

Historically, Russian used fuel from all but VVER-1000 civil reactors has been reprocessed at Mayak to yield RepU with about 0.9% U-235. This has mostly been enriched to provide fuel for RBMK reactors, with the tails as DRepU.

Also historically, to 2000, foreign used fuel was reprocessed and the RepU blended with LEU to yield reactor fuel which was returned as if the RepU had been enriched.

In the centrifuge enrichment process, different ways of feeding cascades with both U_{nat} and RepU and blending the product can control U-232 levels and also U-236 levels (which if over 0.1% can be compensated by higher enrichment levels). Russian enrichment plants have provision for this flexible cascading. Then blending the enriched uranium product (from SIU, DSIU or RepU) with U_{nat} or SIU can further reduce both of these even isotopes according to customer requirements, and below the pending Russian limit of 5 ppb U-232 (now 2 ppb).

This will enable use of RepU in VVER-1000 reactors from 2021 and increase the value of Russian RepU for domestic needs. It will also mean that production and use of RepU are balanced, especially as RBMK units are decommissioned and the Mayak RT-1 plant capacity is increased to 250 t/yr and the PDC at MCC Zheleznogorsk reaches 250 t/yr.

Russia expects to have spare capacity to process foreign RepU from about 2020.

Radioactive waste

Russia's Duma passed a new Federal Law on Radioactive Waste Management in June 2011, after 19 months consideration and many amendments. It was passed by the state Council in July and then signed into law. It establishes a legal framework for radioactive waste management, provides for a national radwaste management system meeting the requirements of the Joint Convention on the Safe Management of Spent Nuclear Fuel and on the Safe Management of Radioactive Waste ratified by Russia in 2006.

In November 2015 the government approved Rosatom’s second federal target programme (FTP NRS-2) for nuclear and radiation safety for 2016 to 2030. "The key issue is the deferred liabilities

accumulated during the 70 years of the nuclear industry, particularly during the time of the Soviet Union." In the first FTP since 2008 Rosatom has completed more than was set out then, against a budget of RUR 123 billion. About 73% of the new FTP budget of RUR 562 billion will be for decommissioning commercial reactors, and the withdrawal of buildings and facilities at Mayak Production Association, Siberian Chemical Combine, Angarsk Electrolysis and Chemical Complex and Novosibirsk Chemical Concentrates Plant – facilities once involved in state defence programmes. Nearly 20% of the funding will go on creating the infrastructure required for the processing and final disposal of used nuclear fuel and radioactive waste; 5% on monitoring and ensuring nuclear and radiation safety; and 2% on scientific and technological support. About 70% of the budget is from federal funds, much of the rest from Rosatom. It will be implemented in three 5-year stages, and involves the transition to new used fuel recycling technologies to close the fuel cycle, establishing a final HLW repository, decommissioning of 82 nuclear & radiation hazardous facilities, two nuclear icebreakers and other tasks.

Rosatom and the National Operator for Radioactive Waste Management – FSUE NO RAO – is responsible for coordination and execution of works associated with radwaste management, notably its disposal. This includes military waste. The law establishes time limits for interim radwaste storage and volume limits for waste generators, and defines how they should bring waste in condition suitable for disposal and transfer it to the national operator along with payment of disposal charges. Import and export of radwaste is banned. All newly-generated waste is the responsibility of its generators who will pay for its disposal and storage, with funds accumulated in the SC Rosatom's bank account as a special fund. However, the 2011 law did not address how to resolve property disputes in siting, nor local authority responsibilities, nor financing mechanisms for affected municipalities. In October 2014 NO RAO submitted to Rosatom proposals for changes in legislation on these matters so that it could proceed with its mandate. In 2015 RUR 6.5 billion will be paid over by various enterprises to Rosatom's reserved fund for radioactive waste disposal, at rates set in 2013 for the period to 2017.

Rosatom plans to draft two more laws: on decommissioning and used fuel management.

FSUE RosRAO is a Moscow-based Rosatom company providing commercial back-end radwaste and decommissioning services for intermediate- and low-level waste as well as handling non-nuclear

radwaste and nuclear decommissioning. It commenced operation in 2009 under a temporary arrangement pending finalisation of regulations under the new legislation, and became part of Rosatom's Life Cycle Back-End Division (LC BED) in 2013. It incorporates Radon, and now has branches in each of seven federal districts. The Kirovo-Chepetsk branch is responsible for decommissioning that conversion plant with 440,000 tonnes of waste by 2025 at a cost of RUR 2.1 billion.

Naval waste

RosRAO's Far East Centre for Radioactive Waste Management is **DalRAO**, near Vladivostok in the Maritime Territory. It has Fokino and Viluchinsk divisions or regions, and operates a long-term open-air storage facility in Razboinik Bay for reactor compartments* from dismantled submarines. The long-term storage facility was under construction from 2006 with Japanese assistance and was commissioned in 2012. It has three nuclear service ships, and the Japanese government donated a floating dock and other equipment to move the reactor compartments. RosRAO plans to have the Regional Center for Conditioning and Long-term Storage of Radioactive Waste (RAW Regional Center) here, mainly for naval waste pending handover to NO RAO. In October 2014 the last spent fuel from dismantled nuclear submarines in the Maritime Territory was dispatched to the Mayak reprocessing plant.

* In 2014 the first three were brought ashore, in 2015 RosRAO planned to move five and then raise the number to ten per year, with a total of 54 three-compartment units to be placed.

RosRAO's Northwest Centre for Radioactive Waste Management is **SevRAO**, in the Murmansk region, which is engaged in remediation of the sites which were Navy Northern Fleet bases, and dismantling of retired nuclear-powered naval ships and submarines as well as nuclear service ships at several sites. Andreeva Bay is the main centre of attention today, and international funding is applied to removing its stock of used naval fuel under the Northern Dimension Environmental Partnership (NDEP), which was established in 2002 and is supported by many countries and the EU through the European Bank for Reconstruction and Development (EBRD). Its Nuclear Window funds work at Andreeva Bay, dismantling *Lepse* and the *Papa*-class submarine at Severodvinsk, with €165 million pledged to mid-2017.

Sayda Bay west of Murmansk was a low-level waste storage site for the navy and has become a regional radioactive waste storage

centre as well as a major ship and submarine dismantling centre. After being docked for 24 years at Atomflot's base near Murmansk, the nuclear service ship *Lepse* was towed to the Nerpa shipyard in Sayda Bay in 2012 and cut up on a slipway over 2013-16, leaving two problematical sections of the hull. It had served as a floating receptacle for used fuel from Russian icebreakers from 1961 to 1988, and stored damaged fuel from the *Lenin*. An aft section contained radioactive waste that was sent to the nearby Sayda Bay facility, and a fore section contained 639 used fuel assemblies from icebreakers, many of them badly damaged, were removed over 2019-21 inside a special structure and sent to Mayak. All this is funded internationally under the NDEP.

The old *Volodarsky*, used as a nuclear service ship from 1966 to 1991 and laden with a lot of low- and intermediate-level radioactive waste, anchored near Murmansk until 2013, was also towed to Sayda Bay, unloaded and then dismantled by the end of 2014. This was funded by the Russian government. Other solid radioactive waste was collected at Andreeva Bay for transport to Sayda Bay for long-term storage. A lot of submarine dismantling was undertaken at Sayda Bay, with many three-compartment reactor units now stored there on land. In August 2021 Rosatom reported that 120 out of 123 decommissioned submarines in the Arctic region had been dismantled.

Gremikha is a current naval base between Murmansk and Archangel where SevRAO is undertaking the defuelling and dismantling of 11 highly-radioactive liquid metal-cooled naval reactors from *Alfa*-class submarines from 2014 to 2023. After the 50-tonne reactors are removed from the hull segments shipped apparently from Sayda Bay, they are put into a hot cell and then defuelled, with the fuel loaded into containers for transport to Mayak for reprocessing. This work takes about a year for each core. Raising the scuttled *K-27* submarine with similar reactors and dismantling it is pending there (see below).

Andreeva Bay, in Litsa Fjord 55 km from the Norway border, was set up in the 1960s as a naval base for nuclear submarine refuelling. In 1982 a major leak from a used fuel pool caused the contents to be transferred to temporary and poorly engineered dry storage. Most of the used fuel from dismantled Northern Fleet submarines was stored at Andreeva Bay – some 22,000 fuel assemblies from 100 naval reactors. In 1992 Norway signed an agreement to address the nuclear legacy issues of the former Northern Fleet and the decommissioning of the nuclear submarines. Andreeva Bay was transferred to civil management in 1993 as Branch #1 of SevRAO. The strategy for removing used fuel from the

original dry storage units was developed from 2002, with funding from the UK. The removal procedure included building an enclosure of the dry storage units, some of which are damaged and leaking, then transferring the fuel to new canisters, which are then put into 40-tonne casks for storage or transport. In May 2014 SevRAO signed a RUR100 million contract with Norway's Finnmark to upgrade the Andreeva Bay dry storage facility, and this was commissioned in 2017. From 2017 to 2020 about 10,000 fuel assemblies were removed from Andreeva Bay to a storage site outside the Murmansk region for disposal.

Submarine fuel is shipped to Andreeva Bay in the 1620 dwt *Rossita*. This is a dedicated ship to transport up to 720 tonnes of used nuclear fuel and radioactive waste, and was built for Atomflot in Italy in 2011. The *Rossita* is primarily for naval waste and fuel from decommissioned submarines, and is used on the Northern Sea Route cruising between Gremikha, Andreeva Bay, Sayda Bay, Severodvinsk and other Russian facilities which dismantle nuclear submarines. *Rossita* also moves casks of used submarine fuel from Andreeva Bay to the railhead at the Atomflot base at Murmansk, for transport to Mayak.

A new vessel built in Italy under a 2013 contract, the semi-submersible pontoon dock *Itarus*, designed to transport three-compartment units of dismantled Russian nuclear submarines for SevRAO in Sayda Bay, was delivered in 2016.

As SevRAO has made good progress, there are plans costed at €123 million to recover seven items of radioactive debris from Arctic waters, where most were dumped in Soviet times, by 2032. This includes submarine reactor compartments and two entire submarines with fuel still in their reactors – *K-27* which was scuttled in 1982 in shallow water after major failure in one of its lead-bismuth cooled reactors, and *K-159* which sank while under tow to decommissioning in 2003. The majority of the debris is in the eastern bays of the Novaya Zemlya, in the Kara Sea. Some is in the Barents Sea. The total radioactivity of nuclear submarines in both seas is estimated at 37 PBq.

Civil waste

RosRAO is envisaged as an international operator, providing back-end fuel cycle services globally.

The National Operator for Radioactive Waste Management (NO RAO) is a federal-state unitary enterprise set up in March 2012 as

the national manager of Russia's used nuclear fuel and radioactive waste, including its disposal. It is the national operator for handling all nuclear waste materials and the single organisation authorised to carry out final disposal of radioactive waste, and also other related functions. Its functions and tariffs are set by government, notably the Ministry of Natural Resources. Its branches are at Zheleznogorsk in Krasnoyarsk, Seversk in Tomsk, Dimitrovgrad in Ulyanovsk and (from late 2013) Novouralsk in Sverdlovsk.

NO RAO is planning an underground research laboratory in Nizhnekansky granitoid massif near Krasnoyarsk for study into the feasibility of disposal of solid HLW and solid medium-level long-lived waste. It has called for tenders, with stage 1 to be completed by the end of 2019, and the whole project completed in 2024. See section below on High-level waste disposal, geological repositories.

The System of State Accounting and Control of Nuclear Materials and Radioactive Waste (SSAC RM&RAW) is intended to perform physical inventory testing of nuclear materials and radioactive waste at their locations, and carry out accounting and control of them at the federal, regional and departmental levels. In February 2015 Rosatom introduced an automated system for accounting and control of radwaste from more than 2000 organisations, which is to be fully implemented by the end of the year.

About 32 million cubic metres of radioactive waste is to be disposed of within the framework of NO RAO's program at a cost of about RUR 307 billion, according to Rosatom. NO RAO's investment program runs to 2035 and includes capital investment in infrastructure of RUR 158 billion ($4.77 billion). Owners of the radioactive waste needing disposal are to provide 80% of that money, while the remaining 20% is to come from the federal budget. In 2013, 24,000 tonnes of used fuel was reported to be awaiting reprocessing or disposal. Rosatom's Social Council plays a major role in achieving public acceptance.

Plant 20 at PA Mayak, Ozersk, is understood to be a military plutonium processing facility employing 1900 people. There was a plan to close it down and transfer operations to the Siberian Chemical Combine at Seversk as part of restructuring the nuclear weapons complex, but this was cancelled in March 2010. In 2011 Rostechnadzor said that urgent attention was needed "to the 20 open liquid radioactive waste pools, including decommissioning those at FGUP PA Mayak as containing the highest concentration and amount of liquid radioactive waste."

Used fuel from Russian-built foreign power and research reactors is repatriated, much of it through the port of Murmansk. Some 70 containers were unloaded and moved south by rail over 2008-2014.

High-level waste disposal, geological repositories

No repository is yet available for high-level waste. Earlier, site selection was proceeding in granite on the Kola Peninsula, and 30 potential disposal sites have been identified in 18 regions, including Siberia, the Urals, the Volga region and the Northwest federal district in order of priority. In 2003 Krasnokamensk in the Chita region 7000 km east of Moscow was suggested as the site for a major spent fuel repository.

Then in 2008 the Nizhnekansky Rock Massif at Zheleznogorsk in Krasnoyarsk Territory was put forward as a site for a national deep geological repository. Rosatom said the terms of reference for the facility construction would be tabled by 2015 to start design activities and set up an underground rock laboratory. Public hearings on the Nizhnekansky Granite Massif were held in July 2012 and in November 2013 it was identified in the Regional Energy Planning Scheme as the planned repository site. In August 2016 the Territorial Planning Scheme to 2030 confirmed the site and approved construction of repository facilities here for 4500 m^3 net of class 1 waste and 155,000 m^3 net of class 2 waste.

The National Operator for Radioactive Waste Management (NO RAO) envisages the establishment of an underground laboratory in the Yeniseysky area near Krasnoyarsk for this waste and then no less than nine years' research. It completed the design documentation for the underground laboratory in March 2015 and expects to begin construction in 2017. A decision on repository construction is due by 2025, and the facility itself is to be completed by 2035. Phase 1 of the facility is to be designed to hold 20,000 tonnes of intermediate- and high-level waste, which will be retrievable.

Low- and intermediate-level waste

These are mostly handled similarly to those in other countries. Radon has been the organisation responsible for medical and industrial radioactive waste. It has had 16 storage sites for waste up to intermediate level. Not far outside Moscow, the major Radon

facility has both laboratories and disposal sites. Other near-surface storage facilities were in 2008 planned for Sosnovy Bor, Glazov, Gatchina, Novovoronezh, Kirovo-chepetsky, Murmansk, Sarov, Saratov, Bilibino, Kransokamensk, Zelenogorsk, Seversk, Dimitrovgrad, Angarsk, and Udomlya.

NO RAO is planning to establish repositories for at least 300,000 m^3 of low- and intermediate-level waste (LILW, class 3&4 radioactive waste), and these plans are to be in place by 2018. One facility would be built in each of Russia's seven federal districts to dispose of these three waste streams. In August 2016 the Territorial Planning Scheme to 2030 approved construction of the following near-surface repository facilities:

- 100,000 m^3 LILW at Ozersk in Chelyabinsk region for Mayak.
- 200,000 m^3 LILW at Tomsk/ Seversk for SCC.
- 48,000 m^3 LILW from Urals Electrochemical Combine at Novouralsk.
- 50,000 m^3 LILW at Sosnovy Bor in the Leningrad oblast.

In December 2015 NO RAO received a licence to operate the first stage of a repository at Novouralsk. The licence permits the near-surface disposal of solid radioactive waste by its Seversk branch on behalf of the Urals Electrochemical Combine, and the first stage of 15,000 m^3 was opened in December 2016. Construction of the second stage is to start in 2017, taking capacity to 54,000 m^3. The facility with a total final capacity of 150,000 m^3 is planned to operate until 2035. "The investments in design, operation and care & maintenance of the facility, as well as subsequent monitoring of the environment will be RUR 6 billion (US$820 million), as per preliminary estimates," according to NO RAO.

NO RAO has received local government approval in the Chelyabinsk and Tomsk regions respectively for the final disposal of low- and intermediate-level waste (LILW) at the sites of Mayak Production Association in Ozersk, and Siberian Chemical Combine (SCC), based in Tomsk. In 2017 NO RAO said it planned a 214,000 m^3 repository near Ozersk, and 150,000 m^3 at Seversk near Tomsk, both to be built by 2021.

However, Russia has also for many years used deep-well injection for low- and intermediate-level waste from some facilities, notably Seversk, Zheleznogorsk and Dimitrovgrad. This is mainly waste from reprocessing. A Central Europe review report in 1999 said that

the wells ranged from 300 up to 1500 metres deep, and that Seversk was the main site utilising the method, with 30 million cubic metres injected. This practice has delayed Russian acceptance of an IAEA standard for radioactive waste disposal, since it has no packaging or engineered barriers and relies on the geology alone for safe isolation. The new 2011 Radioactive Waste Management law said: "Underground disposal of liquid radioactive waste may be executed, in accordance with the requirements of federal regulations and rules, inside geological formations ('collector horizons') as limited by the bounds of the area allotted, within which liquid radioactive waste must remain localised."

In July 2013 Rostechnadzor issued five-year licences to the three regional branches of NO RAO, for "activities associated with final disposal of liquid radioactive waste." In the November 2013 Regional Energy Planning Scheme two active sites for deep geological disposal of liquid radioactive waste (LRW) are identified: Dimitrovgrad, Ulyanovsk oblast, on the NIIAR site 1300 km SE of Moscow, and a northern one: Zheleznogorsk, Krasnoyarsk territory in Siberia, on the MCC site. A preliminary finding of the 2013 IRRS mission from IAEA was that "License conditions related to the safety assessment and safety case of liquid radioactive waste disposal facilities should be revised." In August 2016 the Territorial Planning Scheme to 2030 approved deep well repository for 50 million m^3 of liquid radioactive waste.

Energospetsmontazh announced in March 2015 that the trial operation of plasma-based processing of radioactive waste had started at Novovoronezh. The system is designed for plasma pyrolysis processing of solid radioactive waste of medium and low activity containing both combustible and non-combustible components.

Kyshtym accident and related pollution

There was a major chemical accident at Mayak Chemical Combine (then known as Chelyabinsk-40) near Kyshtym in Russia in 1957. This plant had been built in haste in the late 1940s for military purposes. The failure of the cooling system for a tank storing many tonnes of dissolved nuclear waste resulted in an explosion due to ammonium nitrate having a force estimated at about 75 tonnes of TNT (310 GJ). Most of the 740-800 PBq of radioactive contamination settled out nearby and contributed to the pollution of the Techa River, but a plume containing 80 PBq of radionuclides spread hundreds of kilometres northeast. The affected area was

already very polluted – the Techa River had previously received about 100 PBq of deliberately dumped waste, and Lake Karachay had received some 4000 PBq. This 'Kyshtym accident' killed perhaps 200 people and the radioactive plume affected thousands more as it deposited particularly Cs-127 and Sr-90. It is rated as a level 6 'serious accident' on the International Nuclear Event Scale, only surpassed by Chernobyl and Fukushima accidents.

Up to 1951 the Mayak plant had dumped its waste into the Techa River, whose waters ultimately flow into the Ob River and Arctic Ocean. Then they were disposed of into Lake Karachay until at least 1953, when a storage facility for high-level waste was built – the source of the 1957 accident. Finally, a 1967 duststorm picked up a lot of radioactive material from the dry bed of Lake Karachay and deposited it on to the surrounding province. It appears that some radioactive discharges into the Techa River continued, and that in particular between 2001 and 2004, some 30-40 million cubic metres of radioactive effluent was discharged near the reprocessing facility, which "caused radioactive contamination of the environment with the isotope strontium-90." There is no radiological quantification.

The outcome of these three events made some 26,000 square kilometres the most radioactively-polluted area on Earth by some estimates, comparable with Chernobyl.

Decommissioning

Rostechnadzor oversees a major programme of decommissioning old fuel cycle facilities, financed under the Federal target program on Nuclear and Radiation Safety. The government said it planned to spend some $5 billion to 2015 on decommissioning and waste management. Since 1995 nuclear power plants have contributed to a decommissioning fund.

Several civil reactors are being decommissioned: an experimental 50 MWt LWGR type at Obninsk which started up in 1954 (5 MWe) and was the forerunner of RBMKs, two early and small prototype LWGR (AMB-100 & 200) units – Beloyarsk 1&2 – the Melekess VK-50 prototype BWR, and three larger prototype VVER-440 units at Novovoronezh, a V-210 and V-365 and a V-179. Five were shut down 1981-90 and await dismantling. The fuel has been removed from these and that from Novovoronezh has been shipped to centralised storage in Zheleznogorsk and will be stored there for about ten years before reprocessing. The Beloyarsk fuel is still

onsite since reprocessing technology for it is not yet available. The plant is being dismantled, and the site is due to be clear by 2032.

Shutdown Civil Power Reactors

Reactor	Power, MWe	(Proto)type	Started	Shutdown
Obninsk AM-1	6 (50 MWt)	LWGR	1954	2002
Beloyarsk 1, AMB-100	108	LWGR	1964	1981
Beloyarsk 2, AMB-200	160	LWGR	1968	1990
Melekess	50	VK-50	1964	1988
Novovoronezh 1	210	VVER-440/V-210	1964	1988
Novovoronezh 2	336	VVER-440/V-365	1970	1990
Novovoronezh 3	385	VVER-440/V-179	1971	2016
Leningrad 1	950	RBMK	1974	2018
Bilibino 1	11	LWGR	1974	2019
Leningrad 2	950	RBMK	1976	2020

At Novovoronezh 1&2 a decommissioning project with partial dismantling of equipment was largely completed in 2020. The work will take several years, and buildings are likely to be re-used. In particular that portion of the site houses the district heating pumps and equipment, which provides 75% of the heat for the city, and a spare parts store for Rosenergoatom. Novovoronezh 3 was shut down in December 2016 and it will be cannibalised to keep unit 4 (also V-179) operating for up to 60 years.

In 2010 Siberian Chemical Combine (SCC) in collaboration with Rosatom set up the JSC Pilot Demonstration Center for Decommissioning of Uranium-Graphite Reactors (PDC UGR) at SCC site to implement a decommissioning concept for 13 shut-down uranium-graphite production reactors (PUGR) for military plutonium. These are at Mayak Chemical Combine at Ozersk (5), near Kyshtym, at Siberian Chemical Combine, Seversk (5), and at Mining & Chemical Combine, Zheleznogorsk (3). The last plutonium production reactor, ADE-2 at Zheleznogorsk, finally closed for decommissioning in April 2010.* The fuel has been removed from the shut-down reactors and nearly all of it has been reprocessed at Mayak and Seversk. The concept provides for building multiple safety barriers and sealing of shut-down reactors rather than their dismantling, at a cost estimated to be RUR 2 billion (US$ 67 million) each. Entombment is the option selected for EI-2, ADE-4 and ADE-

5 reactors. All 13 are expected o be decommissioned by 2030. EI-2, also described as Russia's first industrial nuclear power station since it produced power as well as military plutonium, operated to the end of 1990 and was decommissioned in 2015. In 2009 SCC won a tender to prepare for decommissioning of the four Bilibino reactors (due to close 2019-21) and two closed ones at Beloyarsk (all LWGRs).

*Russia's plutonium was produced by 13 reactors at three sites: PO Mayak in Ozersk, also known as Chelyabinsk-65 (A, AV-1-3, AI-IR); SKhK – the Siberian Chemical Combine in Seversk, also known as Tomsk-7 (ADE-3,4&5, EI-1, EI-2); and GKhK – the Mining and Chemical Combine in Zheleznogorsk, also known as Krasnoyarsk-26 (AD, ADE-1&2). The five Mayak reactors produced an estimated 31t of weapons-grade plutonium between 1948 and 1990, the five SKhK reactors produced 68t between 1955 and 2008, and the three GKhK reactors produced 46t between 1958 and 2010. Ten of these reactors were shut down between 1987 and 1992, leaving only ADE-2, 4 and 5 until 2008 & 2010. Of four heavy water reactors at Mayak (OK-180, OK-190, OK-190M and LF-2) the first was intended for plutonium production but in fact all were used for producing isotopes and tritium. LF-2 remains in operation.

In January 2014 Rosatom announced that the PDC UGR, having established its credibility and expertise, would cease to be part of SCC and become part of its new End-of-Life (EOL) Management Division, under the Federal Centre for Nuclear and Radiation Safety (FC NRS).

Three nuclear-powered icebreakers have been decommissioned: *Lenin*, *Sibir* and *Arktika*, also the support vessel: *Lepse* which held some used nuclear fuel from the Arctic fleet. *Lepse* was taken out of the water in October 2014 for further dismantling at the Nerpa Shipyard in Murmansk. *Lenin* is being turned into a museum. SevRAO, the northern branch of RosRAO, dismantles nuclear-powered naval vessels at its Sayda Bay site in Murmansk, and Atomflot is considering using it for retired icebreakers.

In 2014 the Angarsk Electrolysis & Chemical Complex (AECC) said that decommissioning of its conversion plant and diffusion enrichment plants would require RUR 20 billion ($500 million). Decommissioning the conversion capacity at Kirovo-Chepetsky Chemical Combine which was shut down in the 1990s is expected to cost RUR 2.1 billion.

Organisation

The State Corporation (SC) Rosatom is a vertically-integrated holding company which took over Russia's nuclear industry in 2007, from the Federal Atomic Energy Agency (FAEA, also known as Rosatom). This had been formed from the Ministry for Atomic Energy (Minatom) in 2004, which had succeeded a Soviet ministry in 1992. The civil parts of the industry, with a history of over 60 years, are consolidated under JSC AtomEnergoProm (AEP).

During 2008 there was a major reorganisation or "privatisation" of nuclear industry entities involving change from Federal State Unitary Enterprises (FSUE) to Joint Stock Companies (JSC), with most or all of the shares held by AtomEnergoProm. By mid August 2008, 38 of 55 civil nuclear FSUEs had been reformed. Some renaming occurred due to new restrictions on the use of "Russia" or derivatives (eg "Ros") in JSC names. In mid 2014 eight of the remaining FSUEs were designated 'federal nuclear organisation', including Mayak PA and MCC.

The State Nuclear Energy Corporation **Rosatom** (as distinct from the earlier Rosatom agency) is a non-profit company set up in 2007 to hold all nuclear assets, including more than 350 companies and organisations, on behalf of the state. In particular, it holds all the shares in the civil holding company AtomEnergoProm (AEP). It took over the functions of the Rosatom agency and works with the Ministries of Industry and Energy (MIE) and of Economic Development and Trade (MEDT) but does not report to any particular ministry. Early in 2012 the government announced that its civil divisions might be privatized, at least to a 49% share in individual entities. The total workforce is over 250,000.

SC Rosatom divisions are:

- Nuclear weapons complex.
- Nuclear & radiation safety and waste.
- Nuclear power – Atomenergoprom, Rosenergoatom.

- Applied and fundamental science, composite materials.
- Atomflot – Arctic fleet of seven nuclear icebreakers and one nuclear merchant ship.

AtomEnergoProm (Atomic Energy Power Corporation, AEP) is the single vertically-integrated state holding company for Russia's nuclear power sector, separate from the military complex. It was set up at the end of 2007 to consolidate the civil activities of Rosatom including uranium production, engineering, design, reactor construction, power generation, isotope production and research institutes in its several branches, but not used fuel reprocessing or disposal facilities. It incorporates more than 80 enterprises operating in all areas of the nuclear fuel cycle. The April 2007 Presidential decree establishing it specifies nuclear materials, which may be owned exclusively by the state, lists Russian legal entities allowed to possess nuclear materials and facilities, existing joint stock companies to be incorporated into Atomenergoprom, and lists federal state unitary enterprises to be corporatized first and incorporated into Atomenergoprom at a later stage. Exclusive state ownership of nuclear materials had been seen as a barrier to competitiveness and other Russian corporate entities will now be allowed to hold civil-grade nuclear materials, under state control.

Entities from Atomenergoprom itself down to various third-level subsidiaries will be joint stock companies eventually. Public investment in the bottom level operations is envisaged – the joint venture between Alstom and Atomenergomash to provide large turbines and generators is cited as an example.

JSC **AtomEnergoProm's** many entities include the following (most are JSCs):

- ARMZ Uranium Holding Co (JSC AtomRedMetZoloto) – uranium production – owns Russian mine assets.
- Uranium One Group (U1 Group) – responsible for all foreign uranium mining, 78.4% owned.
- Techsnabexport (TENEX) – foreign trade in uranium products and services, with North American subsidiary TENAM.
- JSC Enrichment & Conversion Complex.
- TVEL – conversion, enrichment and nuclear fuel fabrication. The BREST-300 reactor is being built by TVEL at SCC Seversk, apparently due to the integration of fuel cycle facilities in the project.
- **ASE Group** is Rosatom's engineering division, accounting for 30% of the global nuclear power plant construction market according to

Rosatom. Most foreign projects are ASE's reponsibility. It now incorporates the following entities:
- Atomproekt, the new name for VNIPIET (All-Russia Science Research and Design Institute of Power Engineering Technology) which since 2013 incorporates St Petersburg Atomenergoproekt (SPbAEP) – design of nuclear power projects, radiochemical plants and waste facilities. From 2015 this is part of the ASE Group.
- Nizhny-Novgorod Atomenergoproekt (NN AEP or NIAEP) – power plant design, from 2012: holding company for ASE. Sometimes then known as NIAEP-ASE, but re-named Atomstroyexport in December 2016. From October 2014 this is the parent company of Moscow JSC Atomenergoproekt (AEP), so the whole entity became the ASE Group (united company NIAEP-ASE-AEP). Then in 2015 Atomproekt was added to it.
- **Atomstroyexport (ASE)** – construction of nuclear plants abroad, merged with NIAEP in 2012. Sometimes known as NIAEP-ASE until re-named Atomstroyexport in December 2016. From the end of 2014, ASE owns all the shares in JSC Atomenergoproekt and 49% of those in NIAEP, taking them over from Atomenergoprom.
- Moscow Atomenergoproekt (AEP) – power plant design, became part of NIAEP-ASE.
- Energospetsmontazh – construction and assembly, also repair of nuclear plants.
- Atomenergomash (AEM) – a group of companies building reactors.
- OKBM Afrikantov (formerly just OKBM – Experimental Design Bureau of Machine-building – Mashinostroyeniya) at Nizhny Novgorod- reactor design and construction.
- OKB Gidropress (Experimental Design Bureau pressurised water – Hydropress) at Podolsk near Moscow – PWR reactor design.
- JSC Rosenergoatom (briefly Energoatom) – responsible for construction and operation of nuclear power generation.
- Rusatom Overseas was established in 2011 to promote Russian nuclear technologies in world markets. After restructuring in May 2015, it is divided into two companies served by Rusatom International Network which runs Rosatom's regional offices around the world, supporting the activities of Rosatom's divisions in foreign markets, seeking new business opportunities and promoting Rosatom's products and services abroad. The two companies are:
• JSC **Rusatom Energy International**, 44% owned by Rosatom and 56% by Atomenergoprom. It manages foreign construction projects and operation of those nuclear power plants as a shareholder in project companies. It is a major shareholder in JSC Akkuyu Nuclear in Turkey and a 34% shareholder in Fennovoima Oy in Finland. The functions of the company include financing, construction on budget and on time, safe and efficient operation of nuclear power plants,

and sale of electricity on foreign markets.
• JSC **Rusatom Overseas Inc**, based in Moscow and responsible for promotion of the integrated offer of nuclear power plant construction projects in international markets. Its key tasks are growth of the overseas orders portfolio of Rosatom companies and retaining the leading positions of Russia in global nuclear market. It is to ensure full back-up of the customer nuclear power programmes at all stages of implementation, including financing, training, localisation of supply chain, fuel supply with take-back of used fuel for reprocessing, and decommissioning.
- Rusatom Overseas Germany (RAOS Germany) in 2016 will take over the international sales and marketing activities of NUKEM Technologies GmbH in the regions outside of the Western European markets, hence bundling all international marketing activities in the nuclear back-end area and high-temperature reactor fuel with Rusatom Overseas.
- Rusatom Service – coordination of servicing nuclear plants abroad, providing "customised solutions for the modernization and operating period extension of VVER-based nuclear power plants".
- Atomenergoremont – maintenance and upgrading of nuclear power plants,
- NUKEM Technologies GmbH is active worldwide in management of radioactive waste and spent fuel, and decommissioning of nuclear facilities. NUKEM Technologies Engineering Services GmbH focuses on engineering. Both are wholly-owned subsidiaries of JSC Atomstroyexport, and from 2016 are apparently part of Rusatom Overseas.
- Research & Development Institute for Power Engineering (NIKIET) at Moscow – power plant design (originally: submarine power plants)
- Central Design Bureau for Marine Engineering (CDBME) of the Russian Shipbuilding Agency – involved in some reactor design.
- JSC State Specialised Design Institute (SSDI or GSPI) was a direct subsidiary of Atomenergoprom set up in 1948 for producing plutonium but now designing SMRs.

Electricity:

JSC **Rosenergoatom** is the only Russian organization primarily acting as a utility operating nuclear power plants. It was established in 1992 and reorganized in 2001 and then in 2008 as an open JSC. From December 2011 JSC Atomenergoprom holds 96% of the shares, and SC Rosatom (which owns

Atomenergoprom) holds 4%. Rosenergoatom owns all nuclear power plants, both operating and under construction.

InterRAO UES was formerly a joint venture of Rosenergoatom and RAO UES, the utility which was broken up in mid 2008. It is now 57.3% owned by Rosatom and focused on electricity generation in areas such as Armenia and the Kaliningrad part of Russia, as the country's exporter and importer of electricity. It has 8 GWe of generating plant of its own and plans to increase this to 30 GWe by 2015, with the Baltic nuclear plant at Kaliningrad as an early priority. It heads a group of over 20 companies located in 14 countries, involving 18 GWe of capacity. Inter RAO-WorleyParsons (IRWP, with Inter RAO 51%) was set up in mid 2010 to work on the transfer of power engineering technology into Inter RAO's market and to promote Inter RAO's projects oversees.

Engineering and general designers:

In July 2008 the St Petersburg, Moscow and Nizhny-Novgorod divisions of Atomernergoproekt were converted to joint stock companies, with all shares held by Atomenergoprom. The first two are engineering companies and general designers of nuclear power plants mainly using VVER reactors developed by Gidropress. By the end of 2015 all the following engineering companies had been consolidated into the ASE Group as Rosatom's engineering division.

Atomproekt at St Petersburg was formed from the 2013 merger of St Petersburg Atomenergoproekt (SPbAEP) with the All-Russia Science Research and Design Institute of Integrated Power Engineering Technology – VNIPIET (established in 1933) to create the country’s largest nuclear power plant design and development company. It has a particular focus on fast reactors as well as VVER. The company supports all stages of the nuclear fuel cycle, from a decision to start a nuclear power plant construction project to decommissioning. On completion of the merger in mid-2014 it became Atomproekt. Earlier, SPbAEP worked closely with Atomstroyexport (ASE) on exported plants. Atomproekt is responsible for Leningrad II plant, Beloyarsk, Baltic, and also the Belarus, Tianwan, Hanhikivi and Paks II plants as export projects.

Atomproekt is also much involved in fuel fabrication and radioactive waste management. It is Russia's sole design company for used nuclear fuel storage facilities. It is closely involved with the Proryv project for closed fuel cycle with fast reactors.

Atomenergoproekt (formerly Moscow AEP) established in 1986 is a major general design and engineering company for nuclear power

plants. It may also function as general contractor. In October 2014 it became a subsidiary of NIAEP-ASE.

Its version of the AES-2006 evolved to the VVER-TOI, which Rosatom says is planned to be standard for new projects in Russia and worldwide. It is general designer of Novovoronezh II, being built by NIAEP-ASE, Kursk II, Smolensk II as well as Kudankulam in India and Akkuyu in Turkey. It has been responsible for Kursk and Smolensk RBMK plants, Novovoronezh I, Balakovo, and the Zaporozhe, Temelin and Bushehr plants.

NIAEP-ASE: Nizhny-Novgorod Engineering Company Atomenergoproekt (NIAEP) set up in 1951 is building plants at Rostov (Volgodonsk) and Kalinin. NIAEP in March 2012 was merged with Atomstroyexport (ASE) to bolster the latter's engineering capability. (Earlier it had linked with ASE to utilize some 1980s VVER equipment not required for Bulgaria's proposed Belene plant, and built it at Kalinin.) NIAEP became a holding company for JSC ASE, but NIAEP-ASE was being used as acronym to late 2014.

Atomstroyexport (ASE), established by merger in 1998, emerged from the reorganisation as a closed joint stock company owned by Atomenergoprom (50.2%) and Gazprombank (49.8%, it is 69% owned by Gazprom). Early in 2009 the Atomenergoprom and related equity was increased to 89.3% by additional share issue, leaving Gazprombank with 10.7%. It was responsible for export of nuclear plants to China, Iran, India and Bulgaria. In 2009 German-based Nukem Technologies GmbH, which specialises in decommissioning, waste management and engineering services, became a 100% subsidiary of Atomstroyexport. In 2012 ASE merged with Nizhny-Novgorod Atomenergoproekt (NN AEP or NIAEP) to form NIAEP-ASE.

Rosatom, through NIAEP-ASE, offers both EPC (engineering, procurement, construction) and BOO (build, own, operate) contracts for overseas nuclear power plant projects, the latter involving at least 25% Rosatom equity. Rosatom offers various kinds of project financing, including attraction of strategic and institutional investors and debt financing. Some project finance is covered by international agreements involving either export credits, Russian government credit or the participation of Russian state banks. It says that lending rates can be optimized for nuclear power plant projects, and up to 85% of the finance may be provided by government credit from Russia.

In November 2014 the projects in hand on the company website were: Rostov 3&4, Baltic 1&2, Nizhny Novgorod 1&2, Kursk II, all in Russia, and Kudankulam 1&2, Tianwan 3&4, Akkuyu 1-4, Ostrovets 1&2, Bushehr 1, Ninh Thuan 1&2. In mid-2013 Rooppur in Bangladesh was added (but then removed). It is also building a large (3x400 MWe) gas combined-cycle plant: South Ural/Yuzhnouralskaya GRES-2 units 1&2.

NIAEP (post 2012 merger) has a design institute in Nizhny-Novgorod, project management offices in Nizhny-Novgorod, Moscow and St Petersburg, and 11 representative offices in Europe and Asia to oversee projects.

Titan-2 was a major subcontractor for the Leningrad II construction, and in 2015 it took over as general contractor for units 1&2. It will also be general contractor for Hanhikivi in Finland.

Other:

Rusatom Service was set up in October 2011 by Rosenergoatom (51%), Atomenergomash (16%), Gidropress (16%) and Atomtekhenergo (16%). It will undertake maintenance and repair as well as modernization of Russian-design nuclear power plants abroad, applying Russian domestic experience. The company is also to work in the area of technical consultancy, training and retraining of plant personnel. The market is estimated at €1.5 billion per year, rising to €2.5 billion by 2020, including western-design reactors by then.

OTsKS – Rosatom Branch Centre for Capital Construction – was set up in August 2012 to manage its capital investment program in Russia and internationally. It oversees regulatory, technical and legal aspects of capital construction projects, as well as estimating costs and developing schedules. It also provides training for customer-contractors and general contractors such as NIAEP-ASE as well as the personnel of construction companies. Rosatom subsidiary companies had to complete their transition to new rules on planning capital construction projects developed by OTsKS, by the end of 2013. Its main customer is Rosenergoatom which is building about ten units in Russia, with 12 more planned by 2025.

AKME-engineering was established in 2009 to implement the SVBR-100 project at Dimitrovgrad, including design, construction and commercial operation. It is a JV of Rosatom and JSC Irkutskenergo, and is licensed for construction and operation of nuclear plants by Rostechnadzor.

Uralenergostroy in Yekaterinburg is a civil works general contractor responsible for BN-800, BN-1200 and MBIR plants.

The **Federal Centre of Nuclear and Radiation Safety** (FC NRS) is a federal-state unitary enterprise set up in 2007 by Rosatom as part of its End-of-Life (EOL) Management Division. The Pilot Demonstration Center for Decommissioning of Uranium-Graphite Reactors (PDC UGR) is to become part of it, rather than staying with SCC.

The **National Operator for Radioactive Waste Management** (NO RAO) is a federal-state unitary enterprise set up in 2012 responsible for waste management and disposal. It is the National Operator for handling all nuclear waste materials, with functions and tariffs set by government.

FSUE **RosRAO** provides commercial back-end radwaste and decommissioning services for intermediate- and low-level waste as well as handling non-nuclear radwaste. It commenced operation in 2009 under a temporary arrangement pending finalisation of regulations under the new legislation. It incorporates Radon, which was the organisation responsible for medical and industrial radioactive waste, and now has branches in each of seven federal districts. RosRAO's Far East Centre (DalRAO) operates long-term storage for over 70 submarine reactor compartments, pending their recycling. Its northern centre is SevRAO, in the Murmansk region, is engaged in remediation of the sites of Navy Northern Fleet bases, and dismantling of retired nuclear-powered naval ships and submarines. RosRAO is envisaged as an international operator. RosRAO became part of Rosatom's Life Cycle Back-End Division (LC BED) in 2013.

In 2013 Rosatom's **Life Cycle Back-End Division** (LC BED) was set up to incorporate entities hitherto the responsibility of FC NRS: the Mining and Chemical Combine (MCC), RosRAO, SPA V.G.Khlopin Radium Institute and Radon. FC NRS will continue involvement with the new division.

FSUE **Atomflot** is a Rosatom division operating the nuclear powered icebreakers and merchant ship in Arctic waters.

Situation and Crisis Centre of Rosatom was established in 1998 acts as the Operator of the Nuclear Industry System for Prevention and Management of Emergencies. It keeps track of nuclear enterprises and transport of nuclear materials.

SNIIP Systematom is an engineering company for nuclear and radiation safety systems. It will supply the equipment for

automated radiation monitoring systems (ARMS) at the Kalinin 1 nuclear unit in Russia and Tianwan 4 in China.

The VI Lenin All-Russian Electrotechnical Institute and its affiliated Experimental Plant were made FSUEs by presidential decree in March 2015, and removed from the Ministry of Education & Science.

Supply chain entities

Atomenergomash (AEM) was set up in 2006 to control the supply chain for major reactor components. After an equity issue in 2009 it was 63.6% owned by AEP, 14.7% by TVEL and 7.6% by Tenex, and 7% by AEM-finance. In 2009 AEM had sales of RUR 16 billion. AEM companies claim to have provided equipment in 13% of nuclear plants worldwide. Rosatom has one of the largest procurement budgets in the Russian economy, with the annual value of its orders totaling more than RUR 1000 billion ($17.8 billion) in recent years. Almost 85,000 companies are registered as suppliers to Rosatom and 70,000 contracts are signed each year by the group.

Supply chain reliability for nuclear procurement is a significant concern for Rosatom, and it is seeking reform from the Federal Antimonopoly Service (FAS), in particular to ensure a credible ability to deliver high quality goods and services on time rather than just accepting the lowest price. Rosatom wants to conduct audit checks of suppliers prior to their participation in competitive bidding procedures, in order to verify that they would actually be able to fulfil the orders on which they bid. Rosatom cited as an example of the need for procurement reform the purchase of circulation pumps and combined valves for the Novovoronezh power plant. The supplier agreed to a schedule, but this stretched to 80 months and the equipment eventually delivered failed safety tests at the plant. A similar situation occurred at the Beloyarsk plant. The costs of such delays to Rosatom far exceed any compensation it can claim from delinquent suppliers.

The former main nuclear fabrication company, Atommash, was established in 1973 at Volgodonsk and went bankrupt in 1995. It was then profoundly restructured and resurrected as EMK-Atommash before becoming part of JSC Energomash, a major diversified engineering company apparently independent of Rosatom/AEP. Atommash largely moved away from nuclear equipment, though Atomenergomash (subsidiary of AEP) was keen to resuscitate it as an alternative heavy equipment supplier to OMZ. In 2009 Atomenergomash was doing due diligence on the

Energomash group, with a view to taking a half share in it, "to create competition in the segment of monopoly suppliers of long-lead nuclear equipment." In October 2014 AEM-Assets, a subsidiary of Rosatom, acquired the production assets and a 100% interest in Energomash LLC (Volgodonsk)-Atommash, the forging company, and Energomash JSC (Volgodonsk)-Atommash, which provides services related to the lease of equipment and immovable property. Atommash was integrated into Rosatom as part of AEM-Technology, and can now produce four complete sets of nuclear island equipment per year. The reactor pressure vessel supplied to Belarus in 2015 was the first it had produced in 30 years. Two reactor pressure vessels for the RITM-200 reactors for Russia's new icebreaker were also produced in 2015. In 2017 it was building the reactor pressure vessel for the MBIR fast research reactor.

Objedinennye Mashinostroitelnye Zavody (OMZ – Uralmash-Izhora Group) itself is the largest heavy industry company in Russia, and has a wide shareholding. Izhorskiye Zavody, the country's main reactor component supplier, became part of the company in 1999, and Skoda Steel and Skoda JS in Czech Republic joined in 2003. OMZ is expected to produce the forgings for all new domestic AES-2006 model VVER-1200 nuclear reactors (four per year from 2016), plus exports. At present Izhora can produce the heavy forgings required for Russia's VVER-1000 reactors at the rate of two per year, and it is manufacturing components for the first two Leningrad II VVER-1200 units.

The Power Machines Company (JSC Silovye Mashiny Concern, or Silmash) was established in 2000 and brought together a number of older enterprises including Leningradsky Metallichesky Zavod (LMZ), Elektrosila, Turbine Blades Factory, etc. Siemens holds 26% of the stock. Silmash makes steam turbines up to 1200 MWe, including the 1000 MWe turbines for Atomstroyexport projects in China, India and Iran, and has supplied equipment to 57 countries worldwide. It is making 1200 MWe turbine generators for the Leningrad and Novovoronezh II nuclear plants. A significant amount of Power Machines' business is in Asia.

The Russian EnergyMachineBuilding Company (REMCO) was established as a closed joint stock company in Russia in 2008, amalgamating some smaller firms, with half the shares owned by Atomenergomash. It is one of the largest manufacturers of complex heat-exchange equipment for nuclear and thermal power plants, oil and gas industry. Its subsidiaries include JSC Machine-Building Plant ZiO-Podolsk and JSC Engineering Company ZIOMAR.

JSC Machine Building Plant ZiO-Podolsk is one of the largest manufacturers designing and producing equipment for nuclear power and other plants. It has made equipment, including steam generators and heat exchangers, for all nuclear plants in the former USSR. It is increasing capacity to four nuclear equipment sets per year. It appears to be 51% owned by REMCO. It is making the reactor pressure vessel and other main equipment for the BN-800 fast reactor at Beloyarsk as well as steam generators for Novovoronezh, Kalinin 4, Leningrad and Belene.

In April 2007 a joint venture company to manufacture the turbine and generator portions of new nuclear power plants was announced by French engineering group Alstom and JSC Atomenergomash. The 49:51 Alstom-Atomenergomash LLC (AAEM) joint venture, in which both parties would invest EUR 200 million, was established at Podolsk, near Moscow. It includes the technology transfer of Alstom's state of the art Arabelle steam turbine and generator (available up to 1800 MWe) tailored to Russian VVER technology. In 2010 AAEM signed an agreement with Inter RAO-Worley Parsons (IRWP) to establish an engineering consortium to design turbine islands for Russia's VVER reactor-based nuclear power plants. At the same time Alstom signed strategic agreements with major Russian energy companies to jointly provide power generation products and services for Russia's power industry in hydro, nuclear and thermal power generation and electricity transmission. Another agreement, between Alstom Power and Rosatom, details plans to set up a local facility to manufacture Alstom's Arabelle steam turbines for nuclear plants. In 2011 Petrozavodskmash joined the group, and its site is more suitable for shipping large components, so in 2011 the company decided to build its factory for Arabelle manufacture at Petrozavodsk, in Karelia, by 2015 instead of continuing with ZiO-Podolsk near Moscow. First production was expected in 2013 with output reaching three 1200 MWe turbine and generator sets per year in 2016. The Baltic plant will be the first customer, in a RUB 35 billion order, with Russian content about 50%. This will increase to over 70% for subsequent projects.

In September 2007 Mitsubishi Heavy Industries (MHI) signed an agreement with Russia's Ural Turbine Works (UTZ) to manufacture, supply and service gas and steam turbines in the Russian market. Under the agreement, MHI, Japan's biggest machinery maker, will license its manufacturing technologies for large gas turbines and steam turbines to UTZ – part of the Renova Group. The agreement also calls for a joint venture to be established in Russia to provide after-sales service.

Russia has developed several generations of centrifuges for uranium enrichment. Ninth-generation machines are now being deployed, 10th generation ones re being developed, and 11th generation are being designed. The 9th generation units are said to be 1.5 times as efficient as 8th. Overall since 1960, the machine weight, size and power characteristics have remained practically unchanged, but their efficiency was raised more than six-fold, design service life was increased from 3 to 30 years, and the SWU cost was reduced "several times". Centrifuges for China under a US$ 1 billion contract are manufactured at both Tocmash and Kovrov Mechanical plant, both of which will become part of the Fuel Company being established by TVEL. Russia intends to export its centrifuges to the USA and SE Asia.

For more up to date information on heavy engineering, see paper on Heavy Manufacturing of Power Plants.

Early in 2006 Rosenergoatom set up a subsidiary to supply floating nuclear power plants (BNPPs) ranging in size from 70 to 600 MWe. The plants are designed by OKBM in collaboration with others. The pilot plant, now under construction, is 70 MWe plus heat output and incorporates two KLT-40S reactors based on those in icebreakers.

Regulation and safety

Two main laws govern the use of nuclear power: the Federal Law on the Use of Atomic Energy (November 1995 and Federal Law on Radiation Safety of Populations (January 1996). These are supported by federal laws including those on environmental protection (2002) and the Federal Law on Radioactive Waste Management (2011). The 1996 Federal Law on Radiation Safety of Populations is administered by the Federal Ministry of Health.

Rostekhnadzor is the regulator, set up (as GAN) in 1992, reporting direct to the President. Because of the links with military programs, a culture of secrecy pervaded the old Soviet nuclear power industry. After the 1986 Chernobyl accident, changes were made and a nuclear safety committee established. The State Committee for Nuclear and Radiation Safety – Gosatomnadzor (GAN) succeeded this in 1992, being responsible for licensing, regulation and operational safety of all facilities, for safety in transport of nuclear materials, and for nuclear materials accounting. Its inspections can result in legal charges against operators. However, on some occasions when it suspended operating licences in the 1990s,

Minatom successfully overrode this. In 2004 GAN was incorporated into the Federal Ecological, Technological & Atomic Supervisory Service, Rostechnadzor, which has a very wide environmental and safety mandate. It has executive authority for development and implementation of public policy and legal regulation in the environmental field, as well as in the field of technological and nuclear supervision. It controls and supervises natural resources development, industrial safety, nuclear safety (except for weapons), safety of electrical networks, hydraulic structures and industrial explosives. It licences nuclear energy facilities, and supervises nuclear and radiation safety of nuclear and radiologically hazardous installations, including supervision of nuclear materials accounting, control and physical protection. A 2011 overview is on IAEA website.

Safety has evidently been improving at Russian nuclear power plants. In 1993 there were 29 incidents rating level 1 and higher on the INES scale, in 1994 there were nine, and since then to 2003, no more than four. Also, up until 2001 many employees received annual radiation doses of over 20 mSv, but since 2002 very few have done so.

In 2008 Rostechnadzor was transferred to the Ministry of Natural Resources and the Environment, but this was reversed in mid 2010 and it was brought back under direct control of the government and focused on civil nuclear energy. Following other changes in federal legislation, an IAEA Integrated Regulatory Review Service (IRRS) mission in 2013 said that Rostechnadzor had made "significant progress" in its development since 2009 and had “become an effective independent regulator with a professional staff”. Rostechnadzor undertook to make the final IRRS report early in 2014 public.

Glavgosexpertiza, the Russian State Expert Examination Board, is the authority responsible for appraising design documentation and engineering services on behalf of the Ministry of Construction of Russia. Glavgosexpertiza ensures compliance of all major infrastructure construction projects with national technical regulations and statutory requirements.

Rosprirodnadzor, the Federal Service for Supervision of Natural Resources needs to give environmental approval to new projects, through its State Environmental Commission.

Exports: fuel cycle

Soviet exports of enrichment services began in 1973, and Russia has strongly continued this, along with exports of radioisotopes. After 1990, uranium exports began, through Techsnabexport (Tenex). At 2015 Atomexpo it was announced that at the start of the year Rosatom's foreign portfolio totaled US$ 101.4 billion, of which $66 billion was reactors, $21.8 billion was the contracted sales of EUP and SWU, and the remaining $13.6 billion was attributable to the sales of fabricated fuel assemblies and uranium. Rosatom's goal is to gain half its revenue from exported goods and services.

Tenex expects to increase its share in the global market for front-end fuel cycle services to 40% by 2030, assisted by offering an 'integrated product' covering the entire nuclear fuel cycle, and to contribute up to half of Rosatom's foreign currency revenue. Tenex revenue in 2014 was over $2.2 billion, and forward orders totalled almost $23 billion, including almost $6 billion in over 20 contracts with US utilities for enriched uranium product. Tenex sees the Asia-Pacific market as a growth area, using a new transport route through Vostochny Seaport, Primorye Territory.

In 2009 Tenex signed long-term enrichment services contacts with three US utilities – AmerenUE, Luminant and Pacific Gas & Electric – and one in Japan – Chubu. The contracts cover supply from 2014 to 2020. Then it contracted to supply enriched uranium product over the same period with Exelon, the largest US nuclear utility. By the end of 2010, the value of contracts with US companies rose to about $4 billion, beyond the diluted ex-military uranium already being supplied to 2013 from Russian weapons stockpiles. In 2012, Tenex supplied about 45% of world demand for enrichment services and 17% of that for fabricated fuel. It exported fuel for 34 reactors as well as supplying 33 Russian ones.

This US-Russian "Megatonnes to Megawatts" program supplies about 15% of world reactor requirements for enriched uranum and is part of a US$ 12 billion deal in 1994 between US and Russian governments, with a non-proliferation as well as commercial rationale. USEC and Tenex are the executive agents for the program. However, Rosatom confirmed in mid 2006 that no follow-on program of selling Russian high-enriched uranium from military stockpiles was anticipated once this program concludes in 2013. The 20-year program is equivalent to about 140,000 to 150,000 tonnes of natural uranium, and has supplied about half of US needs. By September 2010 it was 80% complete.

TVEL in 2010 won a tender to construct a fuel manufacturing plant in Ukraine, against competition from US company Westinghouse.

Russia's long-term contract to supply fuel to the Ukrainian market is set to run until the end of the useful life of existing Ukrainian reactors, perhaps up to 35 years.

TVEL in 2014 secured contracts with foreign partners that exceeded $3 billion, keeping its ten-year order book at more than $10 billion. Contracts were signed with Finland, Hungary and Slovakia, as well as for research reactors in the Czech Republic, the Netherlands and Uzbekistan. TVEL said it has 17% of the global nuclear fuel supply market.

Rosatom has claimed to be able to undercut world prices for nuclear fuel and services by some 30%.

It was also pushing ahead with plans to store and probably reprocess foreign spent fuel, and earlier the Russian parliament overwhelmingly supported a change in legislation to allow this. The proposal involved some 10% of the world's spent fuel over ten years, or perhaps up to 20,000 tonnes of spent fuel, to raise US$ 20 billion, two thirds of which would be invested in expanding civil nuclear power. In July 2001 President Putin signed into effect three laws including one to allow this import of spent nuclear fuel (essentially an export of services, since Russia would be paid for it).

The President also set up a special commission to approve and oversee any spent fuel accepted, with five members each from the Duma, the Council, the government and presidential nominees, chaired by Dr Zhores Alferov, a parliamentarian, Vice-President of the Russian Academy of Sciences and Nobel Prize physicist. This scheme was progressed in 2005 when the Duma ratified the Vienna Convention on civil liability for nuclear damage. However in July 2006 Rosatom announced it would not proceed with taking any foreign-origin used fuel, and the whole scheme lapsed.

Exports: general, plants and projects

Russia is engaged with international markets in nuclear technology, well beyond its traditional eastern European client states. An important step up in this activity was in August 2011 when Rosatom established Rusatom Overseas company, with authorized capital of RUR 1 billion. In mid-2015 it was split into JSC **Rusatom Overseas** Inc. and JSC **Rusatom Energy International**.

Rusatom Overseas Inc is responsible for implementing non fuel-cycle projects in foreign markets, though apparently it also promotes products, services and technologies of the Russian

nuclear industry generally to the world markets. According to Rosatom, "Rusatom Overseas acts as an integrator of Rosatom's complex solutions in nuclear energy, manages the promotion of the integrated offer and the development of Russian nuclear business abroad, as well as working to create a worldwide network of Rosatom marketing offices." Rusatom Overseas planned to open some 20 offices around the world by 2015, as a market research front and shop window for all Rosatom products and services.

Rusatom Energy International acts "as a developer of Rosatom's foreign projects, which are implemented with the build-own-operate (BOO) structure" and is a shareholder in those project companies. One of the first projects that Rosatom is implementing using the BOO structure is the Akkuyu plant in Turkey. A second project is Hanhikivi in Finland.

At 2015 Atomexpo it was announced that at the start of the year Rosatom's foreign portfolio totaled US$ 101.4 billion, of which $66 billion was reactors, $21.8 billion was the contracted sales of EUP and SWU, and the remaining $13.6 billion was attributable to the sales of fabricated fuel assemblies and uranium. The total at the end of 2015 was over $110 billion, and export revenues in 2015 were $6.4 billion, up 20% from 2014. Rosatom's goal is to gain half its revenue from exported goods and services. Its long-term strategy, approved by its board in late 2011, calls for foreign operations to account for half of its business by 2030. It aims to hold at least one-third of the global enrichment services market by then, as well as 5% of the market for pressurized water reactor (PWR) fuel. The corporation said that it is "actively strengthening its position abroad for the construction of nuclear power plants." In April 2015 Rosatom said that it had contracts for 19 nuclear plants in nine countries, including those under construction (5). In September 2015 it said it had orders for 30 nuclear power reactors in 12 countries, at about $5 billion each to construct, and it was negotiating for 10 more. It said that the total value of all export orders was $300 billion. It aims to have orders for the construction of some 30 power reactors outside of Russia by 2030.

Atomstroyexport (ASE, now NIAEP-ASE) has had three reactor construction projects abroad, all involving VVER-1000 units. It is embarking upon and seeking more, as detailed in Nuclear Power in Russia companion paper, final section on Exports of Nuclear Reactors.

Since 2006 Rosatom has actively pursued nuclear cooperation deals in South Africa, Namibia, Chile and Morocco as well as with

Egypt, Algeria, Jordan, Vietnam, Bangladesh and Kuwait. In 2012 an agreement with Japan was concluded.

Tenex has also entered agreements (now taken over by ARMZ) to mine and explore for uranium in South Africa (with local companies) and Canada (with Cameco).

In September 2008 ARMZ signed a MOU with a South Korean consortium headed by Kepco on strategic cooperation in developing uranium projects. This included joint exploration, mining and sales of natural uranium in the Russian Federation and possibly beyond, but no more has been heard of it.

International collaboration

Russia is engaged with international markets in nuclear energy, well beyond its traditional eastern European client states. In June 2011 Rosatom announced that it was establishing Rusatom Overseas company, a new structure to be responsible for implementing non fuel-cycle projects in foreign markets. It could act as principal contractor and also owner of foreign nuclear capacity under build-own-operate (BOO) arrangements. It is vigorously pursing markets in developing countries and is establishing eight offices abroad.

President Putin's Global Nuclear Infrastructure Initiative was announced early in 2006. This is in line with the International Atomic Energy Agency (IAEA) 2005 proposal for Multilateral Approaches to the Nuclear Fuel Cycle (MNA) and with the US Global Nuclear Energy Partnership (GNEP). The head of Rosatom said that he envisages Russia hosting four types of **international nuclear fuel cycle service centres** (INFCCs) as joint ventures financed by other countries. These would be secure and maybe under IAEA control. The first is an International Uranium Enrichment Centre (IUEC) – one of four or five proposed worldwide (see separate section). The second would be for reprocessing and storage of used nuclear fuel. The third would deal with training and certification of personnel, especially for emerging nuclear states. In this context there is a need for harmonized international standards, uniform safeguards and joint international centers. The fourth would be for R&D and to integrate new scientific achievements.

In March 2008 AtomEnergoProm signed a general framework agreement with Japan's Toshiba Corporation to explore collaboration in the civil nuclear power business. The Toshiba partnership is expected to include cooperation in areas including

design and engineering for new nuclear power plants, manufacturing and maintenance of large equipment, and "front-end civilian nuclear fuel cycle business". In particular the construction of an advanced Russian centrifuge enrichment plant in Japan is envisaged, also possibly one in the USA. The companies say that the "complementary relations" could lead to the establishment of a strategic partnership. Toshiba owns 77% of US reactor builder Westinghouse and is also involved with other reactor technology.

Regarding reactor design, Rosatom has said it is keen to be involved in international projects for Generation IV reactor development and is keen to have international participation in fast neutron reactor development, as well as joint proposals for MOX fuel fabrication.

In April 2007 Red Star, a government-owned design bureau, and US company Thorium Power (now Lightbridge Corporation) agreed to collaborate on testing Lightbridge's seed and blanket fuel assemblies at the Kurchatov Institute with a view to using thorium-plutonium fuel in VVER-1000 reactors, partly in order to dispose of surplus military plutonium (see information papers on Fuel Fabrication and Military Warheads as a Source of Nuclear Fuel for details).

In 2006 the former working relationship with Kazakhstan in nuclear fuel supplies was rebuilt. Kazatomprom has agreed to a major long-term program of strategic cooperation with Russia in uranium and nuclear fuel supply, as well as development of small reactors, effectively reuniting the two countries' interests in future exports of nuclear fuel to China, Japan, Korea, the USA and Western Europe.

In June 2010 Rosatom signed a major framework agreement with the French Atomic Energy Commission (CEA) covering "nuclear energy development strategy, nuclear fuel cycle, development of next-generation reactors, future gas coolant reactor systems, radiation safety and nuclear material safety, prevention and emergency measures." Much of the collaboration will be focused on reprocessing and waste, also sodium-cooled fast reactors. Subsequently EdF and Rosatom signed a further cooperation agreement covering R&D, nuclear fuel, and nuclear power plants - both existing and under construction.

In March 2007 Russia signed a cooperation declaration with the OECD's Nuclear Energy Agency (NEA), so that Russia became a regular observer in all NEA standing technical committees, bringing it much more into the mainstream of world nuclear industry

development. Russia had been participating for some years in the NEA's work on reactor safety and nuclear regulation and is hosting an NEA project on reactor vessel melt-through. This agreement was expected to assist Russia's integration into the OECD, and in October 2011 Russia made an official request to join the NEA. It was accepted as the 31st member of the OECD NEA in May 2012, effective from January 2013. Russia will be represented by its Ministry of Foreign Affairs, Rosatom, and nuclear regulator Rostechnadzor.

Over two decades to about 2010 a Russian-US coordinating committee* was discussing building a GT-MHR prototype at Seversk, primarily for weapons plutonium disposition. Today OKBM is responsible to collaboration with China on HTR development, though NIIAR and Kurchatov Institute are also involved.

* involving SC Rosatom, NIIAR, OKBM, RRC Kurchatov Institute and VNIINM on the Russian side and NNSA, General Atomics, Oak Ridge National Laboratory on the US side.

Research & development

In mid-2009 the Russian government said that it would provide more than RUR 120 billion (about US$3.89 billion) over 2010 to 2012 for a new program devoted to R&D on the next generation of nuclear power plants. It identified three priorities for the nuclear industry: improving the performance of light water reactors over the next two or three years, developing a closed fuel cycle based on deployment of fast reactors in the medium term, and developing nuclear fusion over the long term. Rosatom said that its 2014 spending on R&D would amount to RUR 27-28 billion (US$ 528 million), about 4.5% of its revenue. In 2013 it spent RUR 24 billion, and in 2012 RUR 22.7 billion on R&D. In 2015 Rosatom said that it invested 5% of its revenues in R&D “to reinforce our technological leadership.”

Many research reactors were constructed in the 1950s and 60s. In 2015, 52 non-military research and test reactors were operational in Russia, plus about three in former Soviet republics and eight Russian ones elsewhere. Most of these use ceramic fuel enriched to 36% or 90% U-235. Overall over 130 research reactors have been built based on Russian technology. MBIR is now under construction at Dimitrovgrad.

Kurchatov Institute

Russia has had substantial R&D on nuclear power for seven decades. The premier establishment for this is the Russian Research Centre Kurchatov Institute in Moscow, set up 1943 as the Laboratory No. 2 of the Soviet Academy of Sciences. In 2010 it joined the Skolkovo project, an R&D centre set up to rival Silicon Valley in the USA, and became a Federal State Unitary Enterprise. It has run twelve research reactors there, six of which are now shut down. The 24 kW F-1 research reactor was started up in December 1946 and has passed its 70th anniversary in operation. The largest reactor is IR-8, of 8 MWt, a high-flux unit used for isotope production.

The Kurchatov Institute has designed nuclear reactors for marine and space applications, and continues research on HTRs. Since 1995 it has been involved internationally with accounting, control and physical protection of nuclear materials. US Lightbridge Corporation's seed and blanket fuel assemblies are being tested there with a view to using thorium-based fuel in VVER-1000 reactors.

Kurchatov's Molten Salt Actinide Recycler and Transmuter (MOSART) is fuelled only by transuranic fluorides from uranium and MOX LWR used fuel, without U or Th support. The 2400 MWt reactor has a homogeneous core of Li-Na-Be or Li-Be fluorides without graphite moderator and has reduced reprocessing compared with the original US design. Thorium may also be used, though MOSART is described as a burner-converter rather than a breeder.

Since 1955 the Institute has hosted the main experimental work on plasma physics and nuclear fusion, and the first tokamak systems were developed there. Since 1990, much of its funding comes from international cooperation and commercial projects.

Petersburg Nuclear Physics Institute (PNPI)

The Petersburg Nuclear Physics Institute (PNPI) is near St Petersburg but part of the Kurchatov Institute. It was formerly the B.P. Konstantinov Petersburg Nuclear Physics Institute (PIYaF). In 1959 the 18 MWt WWR-M high-flux research reactor was put into operation, and in 1970 the 1 GeV proton synchrocyclotron SC-1000 started up, these continue in operation.

A 100 MWt high-flux reactor with 25 associated research facilities, PIK, achieved criticality in 2011 at Gatchina but further major work led to its launch at 100 kW in 2019. It uses 27 kg of 90% enriched uranium fuel, tenders for which were called in 2020. PIK is the most powerful high-flux research beam reactor in Russia and is planned to be the basis for the International Centre for Neutron Research. In October 2020 Glavgosexpertiza approved a project for the modernisation of the PIK reactor, and a further launch was announced in February 2021.

The Institute for High Energy Physics and the Institute of Theoretical and Experimental Physics are also part of the Kurchatov Institute, as are the 'Prometheus' Central Research Institute of Structural Materials and the Research Institute of Chemical Reagents and High Purity Chemicals, which were previously part of the Ministry of Education and Science.

Research Institute of Atomic Reactors (RIAR/NIIAR)

Russia's State Scientific Centre – Research Institute of Atomic Reactors (RIAR, or NIIAR) – said to be the biggest nuclear research centre in Russia, is in Dimitrovgrad (Melekess), in Ulyanovsk county 1300 km SE of Moscow. It was founded in 1956 to host both research and experimental reactors, and it researches fuel cycle, radiochemicals and radioactive waste management, as well as producing radionuclides for medicine and industry. It hosts the main R&D on electrometallurgical pyroprocessing, especially for fast reactors, and associated vibropacked fuel technology for these.

RIAR/NIIAR has the largest materials study laboratory in Eurasia, used particularly for irradiated fuel.* The complex's major future role will be in fuel reprocessing. The initial fuel for MBIR is likely to be from reprocessed BOR-60 fuel, as also intended for SVBR-100. In 2014 construction of a new multifunctional radiochemical research centre for closed fuel cycles for fast reactors commenced as part of the revised federal target programme for 2010-2015 and until 2020. Fuel research at RIAR already includes integration of minor actinides into FNR closed fuel cycle, nitride fuel (both mononitride and U-Pu nitride), metallic fuel (U-Pu-Zr, U-Al, U-Be) and RBMK spent fuel conditioning. It also is working on molten salt fuel – reprocessing and minor actinide behaviour, though Kurchatov Institute seems to be the main locus of MSR research.

* In 2010 TerraPower from the USA proposed that RIAR should carry out in-pile tests and post-irradiation examinations of structural materials and fuel specimens planned for its travelling-

wave reactor. A final agreement was expected in November, but apparently did not eventuate.

RIAR's first research reactor – **SM** – has been running since 1961 and now produces radioisotopes and does materials testing. It is a 100 MWt very high-flux water-cooled pressure vessel-type reactor originally using 90% enriched fuel with a neutron trap that operates in the intermediate neutron spectrum. It has been modernised several times and as SM-3 it was recommissioned in 1993. In 2020 it again had a new core. It is expected to operate until 2040.

The **MIR-MR** loop-type reactor commissioned in 1967 is used for testing fuels in runs up to 40 days at up to 100 MWt. It has been important in developing fuel rod designs for power and naval reactors. It is testing the first batch of REMIX fuel and also accident-tolerant fuel (ATF). It has a beryllium moderator and uses 90% enriched fuel. It was due to be retired in 2020.

The small pool-type reactors **RBT-6** & **RBT-10/2** commissioned in 1975 and 1984 are used for long-term experiments and use the spent fuel assemblies from SM. They are 6 & 7 MWt respectively.

As well as three other research reactors, the **BOR-60*** experimental fast reactor is operated here by RIAR – the world's only operating fast research reactor. It started up in 1969 and is to be replaced with the **MBIR**, with four times the irradiation capacity.

* BOR = bystry opytniy reaktor. BOR-60 was licensed to 2015 but was extended to December 2020.

The **multi-purpose fast neutron research reactor – MBIR*** – will be a 150 MWt multi-loop reactor capable of testing lead or lead-bismuth and gas coolants as well as sodium, simultaneously in three parallel outside loops. Initially it will have sodium coolant. It will run on vibropacked MOX fuel with plutonium content of 38%, produced at RIAR in existing facilities. A 24% Pu fuel may also be used. RIAR intends to set up an on-site closed fuel cycle for it, using pyrochemical reprocessing it has developed at pilot scale. MBIR's cost was estimated at RUR 40 billion in 2015. Rostechnadzor granted a site licence to RIAR in August 2014, and a construction licence in May 2015. Construction started in September 2015. Completion was expected in 2020, but the project was paused after starting construction. In November 2020 Rosatom appointed a new contractor, AO Institut Orgenergostroy, and construction resumed, with commissioning expected in 2028. The reactor pressure vessel is being made by Atommash at Volgodonsk.

* MBIR = mnogotselevoy issledovatilskiy reaktor na bystrych neytronach.

Russia's only boiling water reactor, the prototype VK-50 of 200 MWt was commissioned in 1964 and was due to be retired in 2020.

Rosatom is setting up an **International Research Centre** (IRC) based on MBIR and is inviting international participation in connection with the IAEA INPRO programme. In June 2013 an agreement with France and the USA was signed to this end. In April 2017 Rosatom was soliciting Japanese involvement. The full MBIR research complex is now budgeted at $1 billion, with the Russian budget already having provided $300 million from the federal target programme. Pre-construction shares of 1% were being offered for $10 million, allowing involvement in detailed design of irradiation facilities. From 2020 the fee would rise to $36 million per 1% share. RIAR will be the legal owner of MBIR, performing operational and administrative functions, while the International Research Centre will be the legal entity responsible for marketing and research management. In May 2017 Rosatom announced that the multifunctional radiochemical research facility under construction at RIAR would be included in the IRC, to be used for testing technologies to close the fast reactor fuel cycle.

The first 100 MWe **Lead-Bismuth Fast Reactor (SVBR)** from Gidropress was to be built at RIAR, but the project was dropped in 2018. It was designed to use a wide variety of fuels, though the demonstration unit would initially have used uranium enriched to 16.3%. With U-Pu MOX fuel it would operate in closed cycle. It was described by Gidropress as a multi-function reactor, for power, heat or desalination.

RIAR has established a joint venture with JSC Izotop – Izotop-NIIAR – to produce Mo-99 at Dimitrovgrad from 2010, using newly-installed German equipment. This aimed to capture 20% of the world market for Mo-99 by 2012, and 40% subsequently. In September 2010 JSC Isotop signed a framework agreement with Canada-based MDS Nordion to explore commercial opportunities outside Russia on the basis of this JV, initially over ten years.

Institute of Physics and Power Engineering (FEI/IPPE)

In 1954 the world's first nuclear powered electricity generator began operation in the then closed city of Obninsk at the Institute of Physics and Power Engineering (FEI or IPPE). The AM-1* reactor is

water-cooled and graphite-moderated, with a design capacity of 30 MWt or 5 MWe. It was similar in principle to the plutonium production reactors in the closed military cities and served as a prototype for other graphite channel reactor designs including the Chernobyl-type RBMK** reactors. AM-1 produced electricity until 1959 and was used until 2000 as a research facility and for the production of isotopes. FEI also bid to host the MBIR project.

* AM = atom mirny – peaceful atom

** RBMK = reaktor bolshoi moshchnosty kanalny – high power channel reactor

In the 1950s the FEI at Obninsk was also developing fast breeder reactors (FBRs), and in 1955 the BR-1* fast neutron reactor began operating. It produced no power but led directly to the BR-5 which started up in 1959 with a capacity of 5 MWt which was used to do the basic research necessary for designing sodium-cooled FBRs. It was upgraded and modernised in 1973 and then underwent major reconstruction in 1983 to become the BR-10 with a capacity of 8 MWt which is now used to investigate fuel endurance, to study materials and to produce radioisotopes.

* BN = bystry reaktor – fast reactor

Research & Development Institute for Power Engineering (NIKIET)

NIKIET in Moscow is one of Russia's major nuclear design and research centres with a primary focus on advanced reactor technologies including those for regional power supplies, research and isotope production reactors, and neutronic systems for the international fusion reactor (ITER).

NIKIET is at concept development stage with a seabed reactor module – SHELF – a 6 MWe, 28 MWt remotely-operated PWR with low-enriched fuel of UO_2 in aluminium alloy matrix. Fuel cycle is 56 months. The SHELF module uses an integral reactor with forced and natural circulation in the primary circuit, in which the core, steam generator, motor-driven circulation pump and control and protection system drive are housed in a cylindrical pressure vessel. The reactor and turbogenerator are in a cylindrical pod about 15 m long and 8 m diameter, sitting on the sea bed. It is intended as electricity supply for oil and gas developments in Arctic seas. In 2018 NIKIET also proposed its use for the RUR 100 billion Pavlovsky lead-zinc mine project in northern Novaya Zemlya.

In 2010 the government was to allocate RUR 500 million (about US$ 170 million) of federal funds to design a space nuclear propulsion and generation installation in the megawatt power range. In particular, SC Rosatom was to get RUR 430 million and Roskosmos (Russian Federal Space Agency) RUR 70 million to develop it. The work would be undertaken by (NIKIET) in Moscow, based on previous developments including those of nuclear rocket engines. A conceptual design was expected in 2011, with the basic design documentation and engineering design to follow in 2012. Tests were planned for 2018.

Since 2010 NIKIET is also involved with Luch Scientific Production Association (SPA Luch) and a Belarus organization, the Joint Institute for Power Engineering and Nuclear Research (Sosny), to design a small transportable nuclear reactor. The project draws on Sosny's experience in designing the Pamir-630D truck-mounted small nuclear reactor, two of which were built in Belarus from 1976 during the Soviet era. This was a 5000 kWt/630 kWe HTR reactor using 45% enriched fuel in rods with zirconium hydride moderator and driving a gas turbine with dinitrogen tetroxide (N_2O_4) through the Brayton cycle. After some operational experience in 1985-86 the Pamir project was scrapped. The new design will be a similar HTR concept but about 2 MWe.

Joint Institute for Nuclear Research

The Joint Institute for Nuclear Research, at Dubna near Moscow, is an international physics research centre with 18 member states and six associate members. It has the **IBR-2M** fast periodic pulsed reactor of 2 MWt, commissioned in 1984 and modernised in 2010 with higher neutron flux. It uses plutonium oxide fuel.

Mining & Chemical Combine (MCC)

At the Mining & Chemical Combine (MCC), Zheleznogorsk the ADE2 reactor was the third nuclear reactor of its kind built in Russia and came on line in 1964, primarily as a plutonium production unit. However, from 1995 heat and electricity production became its main purposes. The ADE-2 operating experience contributed to technological measures to justify and extend service lives of RBMK reactors at nuclear power plants, with considerable economic benefit and safety improvement. This work was given a governmental science and technology award in 2009. ADE2 was

closed for final decommissioning in April 2010 after "46 years of nearly faultless operation".

MCC Zheleznogorsk also produces granulated MOX for vibropacked FNR fuel, using both military and civil plutonium.

Other R&D establishments

PA Mayak at Ozersk is the main production centre for radioisotopes.

The **Institute for Reactor Materials** (IRM) is at Zarechny, near Beloyarsk, Penza oblast.

TVEL's A.A. Bochvar High Technology Research Institute of Inorganic Materials (**VNIINM**) at Mayak supplies components for fast reactor fuel assemblies. It earlier developed the technology for reprocessing spent uranium-beryllium fuel from liquid metal-cooled fast reactors in dismantled Alpha-class nuclear submarines.

The All-Russian **Scientific and Research Institute for Nuclear Power Plant Operation** (**VNIIAES**) in Moscow was founded in 1979 to provide scientific and technical support for operation of nuclear power plants aimed at improving their safety, reliability and efficiency as well as scientific coordination of the setup of mass-constructed nuclear power facilities.

In 2009 the Moscow Engineering and Physics Institute (MEPhI) was renamed the **National Research Nuclear University** and reformed to incorporate a number of other educational establishments. While partly funded by Rosatom, it is the responsibility of the Federal Education Agency (Rosobrazovaniye).

Public opinion

An April 2008 survey carried out by the Levada Centre found that 72% of Russians were in favour of at least preserving the country's nuclear power capacity and 41% thought that nuclear was the only alternative to oil and gas as they deplete. Over half said that they were indignant about Soviet attempts to cover up news of the Chernobyl accident in 1986.

In April 2010 the Levada Centre polled 1600 adults and found that 37% supported current levels of nuclear power, 37% favoured its active development (making 74% positive), while 10% would like a phase-out and 4.3% would prefer to abandon it completely. 42.6%

saw no alternative to nuclear power for replacing depleting oil and gas.

Immediately after the Fukushima accident in 2011 Levada had only 22% for active development, 30% maintaining current level (ie 52% positive), 27% wanting a phase-out and 12% wanting to abandon it.

In February 2012 a Levada Centre poll showed that 29% of respondents favoured active development of nuclear power, while 37% support retaining it at the current level, so 66% positive. Only 15% of suggested phasing it out, and 7% preferred abandoning nuclear.

The Russian Public Opinion Research Center (VCIOM) took a poll in April 2012 on the anniversary of the Chernobyl accident. It found that 27% of Russians support nuclear power development – up from 16% in 2011, 38 % agree with the present level, and 26% want to reduce it. Nuclear development is supported by young (32%), highly-educated Russians (31%), residents of cities with a population of one million and more, large cities and towns (30-33%). Regarding safety, 35% consider plants of be sufficiently safe, and 57% don't.

In 2015 a poll commissioned by Rosenergoatom found that a clear majority of citizens living near nuclear power plants were in favour of them, and that support had grown since 2013. Most figures for the local plants were more than 70% favourable, and for nuclear power development they were above 80%.

Non-proliferation

Russia is a nuclear weapons state, and a depository state of the Nuclear Non-Proliferation Treaty (NPT) under which a safeguards agreement has been in force since 1985. The Additional Protocol was ratified in 2007. However, Russia takes the view that voluntary application of IAEA safeguards are not meaningful for a nuclear weapons state and so they are not generally applied. One exception is the BN-600 Beloyarsk-3 reactor which is safeguarded so as to give experience of such units to IAEA inspectors.

However, this policy is modified in respect to some uranium imports. All facilities where imported uranium under certain bilateral treaties goes must be on the list of those eligible and open to international inspection, and this overrides the voluntary aspect of voluntary offer agreements. It includes conversion plants, enrichment, fuel

fabrication and nuclear power plants. Also the IUEC at Angarsk will be open to inspection.

Russia undertook nuclear weapons tests from 1949 to 1990.

Russia's last plutonium production reactor which started up in 1964 was finally closed down in April 2010 - delayed because it also provided district heating, and replacement plant for this was ready until then. The reactor may be held in reserve for heating, not dismantled. The other two such production reactors were closed in 2008. All three closures are in accordance with a 2003 US-Russia agreement.

Peaceful Nuclear Explosions

The Soviet Union also used 116 nuclear explosions (81 in Russia) for geological research, creating underground gas storage, boosting oil and gas production and excavating reservoirs and canals. Most were in the 3-10 kiloton range and all occurred 1965-88.

Appendix

Background: Soviet nuclear culture

In the 1950s and 1960s Russia seemed to be taking impressive steps to contest world leadership in civil development of nuclear energy. It had developed two major reactor designs, one from military plutonium production technology (the light water cooled graphite moderated reactor – RBMK), and one from naval propulsion units, very much as in USA (the VVER series - pressurised, water cooled and moderated). An ambitious plant, Atommash, to mass produce the latter design was taking shape near Volgodonsk, construction of numerous nuclear plants was in hand and the country had many skilled nuclear engineers.

But a technological arrogance developed, in the context of an impatient Soviet establishment. Then Atommash sunk into the Volga sediments, Chernobyl tragically vindicated western reactor design criteria, and the political structure which was not up to the task of safely utilising such technology fell apart. Atommash had been set up to produce eight sets of nuclear plant equipment each year (reactor pressure vessels, steam generators, refueling

machines, pressurizers, service machinery – a total of 250 items). In 1981 it manufactured the first VVER-1000 pressure vessel, which was shipped to South Ukraine NPP. Later, its products were supplied to Balakovo, Smolensk (RBMK), and Kalinin in Russia, and Zaporozhe, Rovno and Khmelnitsky plants in Ukraine. By 1986 Atommash had produced 14 pressure vessels (of which five have remained at the factory), instead of the eight per year intended. Then Chernobyl put the whole nuclear industry into a long standby. Russia was disgraced technologically, and this was exacerbated by a series of incidents in its nuclear-propelled navy contrasting with a near-impeccable safety record in the US Navy.

An early indication of the technological carelessness was substantial pollution followed by a major accident at Mayak Chemical Combine (then known as Chelyabinsk-40) near Kyshtym in 1957. The failure of the cooling system for a tank storing many tonnes of dissolved nuclear waste resulted in a non-nuclear explosion having a force estimated at about 75 tonnes of TNT (310 GJ). This killed 200 people and released some 740 PBq of radioactivity, affecting thousands more. Up to 1951 the Mayak plant had dumped its waste into the Techa River, whose waters ultimately flow into the Ob River and Arctic Ocean. Then they were disposed of into Lake Karachay until at least 1953, when a storage facility for high-level waste was built – the source of the 1957 accident. Finally, a 1967 duststorm picked up a lot of radioactive material from the dry bed of Lake Karachay and deposited it on to the surrounding province. The outcome of these three events made some 26,000 square kilometres the most radioactively-polluted area on Earth by some estimates, comparable with Chernobyl.

After Chernobyl there was a significant change of culture in the Russian civil nuclear establishment, at least at the plant level, and this change was even more evident in the countries of eastern Europe who saw the opportunity for technological emancipation from Russia. By the early 1990s a number of western assistance programs were in place which addressed safety issues and helped to alter fundamentally the way things were done in the eastern bloc, including Russia itself. Design and operating deficiencies were tackled, and a safety culture started to emerge. At the same time some R&D programs were suspended.

Both the International Atomic Energy Agency and the World Association of Nuclear Operators contributed strongly to huge gains in safety and reliability of Soviet-era nuclear plants – WANO having come into existence as a result of Chernobyl. In the first two years of WANO's existence, 1989-91, operating staff from every nuclear plant in the former Soviet Union visited plants in the west

on technical exchange, and western personnel visited every FSU plant. A great deal of ongoing plant-to-plant cooperation, and subsequently a voluntary peer review program, grew out of these exchanges.

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